

Anti-infectives

Introduction

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- **Antimicrobial Agent**
 - *Use of chemicals against invading organisms.*
- **Antibiotic**
 - *is a chemical that is produced by one microorganism and has the ability to harm other microbes.*
 - are antibacterial substances produced by various species of microorganisms that **suppress the growth of other microorganisms.**
 - synthetic antimicrobial agents, such as ***sulfonamides*** and ***quinolones***.

Introduction

- ❖ **Antimicrobial agents** provide some of the most dramatic examples of the advances of **modern medicine**
- ❖ Infectious diseases once considered incurable and potentially **lethal** can now be treated effectively with antibiotics
- ❖ The remarkably powerful and specific activity of **antimicrobial drugs** is due to their selectivity for targets

Among these targets are

- bacterial and fungal cell wall-synthesizing enzymes
- the bacterial ribosome
- the enzymes required for nucleotide synthesis and DNA replication
- and the machinery of viral replication

Introduction...

- **Chemotherapeutic Spectra:**
 - **Narrow-spectrum antibiotics:**
 - Acting only on a single or a limited group of microorganisms
 - **Extended-spectrum antibiotics:**
 - Effective against G(+) organisms & also against a significant number of G(-) bacteria
 - **Broad-spectrum antibiotics:**
 - Effective against a wide variety of microbial species

Introduction.....

➤ Antibiotics have three general uses

- Empirical (initial) therapy
 - ✓ Antibiotic should cover all the likely pathogens
 - ✓ Either combination therapy or a single broad-spectrum agent
- Definitive therapy
 - ✓ Once the infecting microorganism is identified,
a
 - narrow-spectrum, low-toxicity agent is instituted
- Prophylactic or preventive therapy

Introduction.....

Classification of antimicrobials

➤ Based on chemical structure and proposed MOA

1. Cell wall synthesis inhibitors , e.g. β -lactam antibiotics,
 - *cycloserine*, *vancomycin*, and *bacitracin*
2. Agents that act directly on the cell membrane of the microorganism, e.g. *Polymyxin*, *nystatin* and *amphotericin B*, daptomycin
2. Agents that affect ribosomal protein synthesis (e.g., *CAF*, *TTCs*, *erythromycin*, *clindamycin*, *aminoglycosides*)
2. Agents that affect bacterial nucleic acid metabolism, such as the rifamycins and the quinolones
2. The antimetabolites, including *trimethoprim* & *sulfonamides*

Introduction

- ❖ Antibiotics are classified as bacteriostatic
- ❖ For **bacteriostatic** drugs, the concentrations that inhibit growth are much lower than those **that kill bacteria**
- ❖ For **bactericidal agents** there is little difference between the **concentrations** that inhibit growth and those that kill bacteria

Introduction...

- ❖ Bactericidal agents can be divided into two groups:
 - agents that exhibit **concentration-dependent** killing (eg, aminoglycosides and quinolones)
 - and agents that **exhibit time-dependent killing** (eg, β -lactams and vancomycin).
- **Minimal inhibitory concentration (MIC):** Lowest concentration of antimicrobial drug capable of inhibiting growth of an organism in a defined growth medium
- **Selective toxicity:** More toxic to the invader than to the host; a property of useful antimicrobial drugs

Table: Bactericidal and bacteriostatic antibacterial agents

Bactericidal Agents	Bacteriostatic Agents
Aminoglycosides	Chloramphenicol
Bacitracin	Clindamycin
β -Lactam antibiotics	Ethambutol
Daptomycin	Macrolides
Fosfomycin	Nitrofurantoin
Glycopeptide antibiotics	Novobiocin
Isoniazid	Oxazolidinones
Ketolides	Sulfonamides
Metronidazole	Tetracyclines
Polymyxins	Tigecycline
Pyrazinamide	Trimethoprim
Quinolones	
Rifampin	
Streptogramins	

Antibacterial Drugs

Cell wall synthesis inhibitors

Beta-lactam Compounds

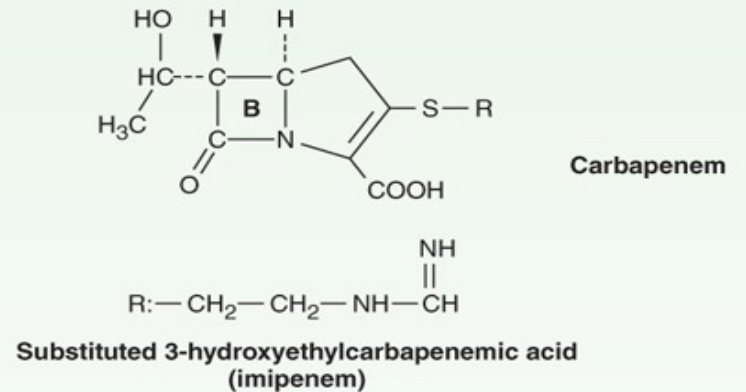
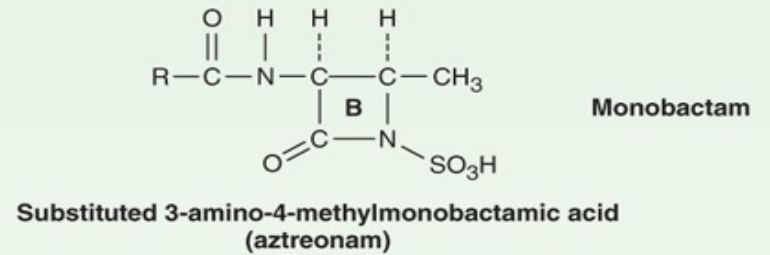
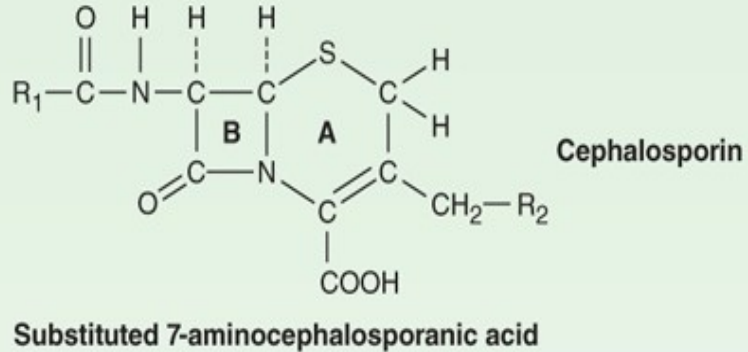
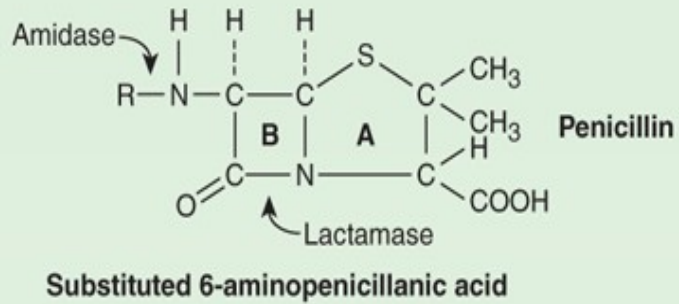
Penicillins

- ❖ The penicillins share features of **chemistry, mechanism of action, pharmacology** and **immunologic** characteristics with cephalosporins.

Chemistry-

- All penicillins have the basic structure. A thiazolidine ring (A) is attached to a β -lactam ring (B) that carries a secondary amino group (RNH–).
- Structural integrity of the **6-aminopenicillanic acid** nucleus (rings A plus B) is essential for the biologic activity of these compounds.
- Hydrolysis of the β -lactam ring by bacterial **β -lactamases** yields **penicilloic acid**, which lacks antibacterial activity.

Chemistry...



Classifications of penicillins

- Substituents of the **6-aminopenicillanic acid** moiety determine the essential **pharmacologic and antibacterial properties** of the resulting molecules.
- Penicillins can be assigned to one of the following **three groups**.

Penicillins (eg, penicillin G)—have

- Greatest activity against **Gram-positive organisms, Gram-negative cocci**, and non- β -lactamase-producing anaerobes.
- little activity against **Gram-negative** rods, and
- they are susceptible to **hydrolysis by β -lactamases**.

Antistaphylococcal penicillins (eg, nafcillin)—resistant to staphylococcal β -lactamases.

- active against **staphylococci** and **streptococci** but not against **enterococci**, **anaerobic** bacteria, and **Gram-negative cocci** and rods.

Extended-spectrum penicillins (aminopenicillins and antipseudomonal penicillins)—

- Active against Gram-negative rods.
- they are relatively susceptible to hydrolysis by β -lactamases.

Mechanism of Action

- ❖ Penicillins, like **all β -lactam** antibiotics, interfering with the **transpeptidation** reaction of bacterial **cell wall synthesis**.
 - Bacterial cell wall is **cross-linked polymer of polysaccharides & pentapeptides**
 - Penicillins Interact with cytoplasmic **Mn binding proteins (PBP)** to inhibit transpeptidation rxns involved in cross-linking the final step in cell wall synthesis

Mechanisms of Penicillin resistance

- ❖ Resistance to penicillins and other β -lactams is due to one of four general mechanisms:
 - Inactivation of antibiotic by β -lactamase,
 - Modification of target pbps,
 - Impaired penetration of drug to target pbps, and
 - Antibiotic efflux.
- ❖ **Beta-lactamase** production is the most common mechanism

Pharmacokinetics

- ❖ Oral Absorption for penicillins differs depending in part on **their acid stability and protein binding**
 - Gastrointestinal absorption of **nafcillin** is erratic, so it is not suitable for oral administration.
 - **Dicloxacillin, ampicillin, and amoxicillin** are **acid-stable** and relatively **well absorbed**.
- ❖ **IV** administration of **penicillin G** is preferred to the IM route because of **irritation and local pain**
- ❖ **Distribution** – **nafcillin** is highly protein-bound penicillins than **Pen-G** and **ampicillin**

Pharmacokinetics...

- ❖ **Benzathine** and **procaine** penicillins are formulated to delay absorption,
- ❖ A single IM injection of **1.2** million units of **benzathine penicillin** maintains serum levels above **0.02 mcg/mL** for 10 days.
- ❖ After 3 weeks, levels still exceed **0.003 mcg/mL**

Enough to prevent most β -hemolytic streptococcal infections.

- ❖ A **600,000**-unit dose of procaine penicillin yields peak concentrations of **1–2 mcg/mL** and clinically useful concentrations for **12–24** hours after a single intramuscular injection.

Pharmacokinetics...

❖ Penicillins

- rapidly excreted by the kidneys;
- small amounts are excreted by other routes.
- Tubular secretion** accounts for about **90% of renal excretion**, and glomerular filtration accounts for the remainder.
- The normal half-life of **penicillin G** is **approximately 30 minutes** but
 - in renal failure, may be as long as **10 hours**.

Pharmacokinetics...

- **Ampicillin and the extended-spectrum penicillins are secreted more slowly than penicillin G and have half-lives of 1 hour.**
- **In renal problems**
 - one-fourth to one-third the normal dose being administered if creatinine clearance is 10 mL/min or less
- **Nafcillin is primarily cleared by biliary excretion.**
- **Oxacillin, dicloxacillin, and cloxacillin are both the kidney and biliary excretion**
 - No dosage adjustment is required for these drugs in **patients in renal failure**

Clinical uses

- ❖ Except for amoxicillin,
 - Oral penicillins should be given 1–2 hours before or after a meal;
- ❖ They should **not be given with food** to minimize binding to food proteins and **acid inactivation**.
- ❖ **Amoxicillin** may be given **without regard to meals**.
- ❖ **Blood levels of all penicillins** can be raised by simultaneous administration of probenecid, 0.5 g (10 mg/kg in children) ...QID

Penicillin

- ❖ Penicillin G is a drug of choice for infections caused by
 - ✓ streptococci,
 - ✓ meningococci,
 - ✓ some enterococci,
 - ✓ penicillin-susceptible pneumococci,
 - ✓ staphylococci confirmed to be non- β -lactamase-producing,
 - ✓ *Treponema pallidum* and certain other spirochetes, some *Clostridium* species,
 - ✓ *Actinomyces* and certain other Gram-positive rods, and
 - ✓ non- β -lactamase-producing Gram-negative anaerobic organisms.

Dose :

- ✓ 4 -24 million units/day per iV in four to six divided doses.
- ✓ High-dose penicillin G can also be given as a continuous intravenous infusion
- ❖ Penicillin V, the oral form of penicillin,.
- ❖ Benzathine penicillin and procaine penicillin G for intramuscular injection yield low but prolonged drug levels.

Penicillins Resistant to Staphylococcal Beta-Lactamase

Dugs: **Methicillin, Nafcillin, and Isoxazolyl Penicillins**

- ✓ for β -lactamase-producing staphylococci
- ✓ An isoxazolyl penicillin such as [dicloxacillin](#), 0.25–0.5 g orally every 4–6 hours (15–25 mg/kg/d for children) **for mild to moderate localized staphylococcal infections.**
- ✓ These drugs are relatively **acid-stable and have reasonable bioavailability.**
- ✓ **food interferes with absorption**, and the drugs should be administered **1 hour before or after meals**

Penicillins Resistant to Staphylococcal Beta-Lactamase

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- ❖ **Oxacillin and nafcillin (IV)... choice for serious staphylococcal infections such as endocarditis.**
- ❖ **Methicillin... no longer use because of ADRs**

Extended-Spectrum Penicillins

Drug: Aminopenicillins, Carboxypenicillins, and Ureidopenicillins

- ❖ greater activity than penicillin G against **Gram-negative bacteria**
 - ✓ Gram-negative outer membrane.
 - ✓ Like penicillin G, they are **inactivated by many β -lactamases**
- ❖ The aminopenicillins, **ampicillin and amoxicillin**, have very similar spectrums of activity
 - ✓ Amoxicillin is better absorbed orally.
 - ✓ Amoxicillin, 250–500 mg three times daily= ampicillin given **QID** times daily

- Both have essentially identical spectrum of antimicrobial action, except that amoxicillin is less effective than ampicillin for shigellosis.

Extended-Spectrum Penicillins (Aminopenicillins, Carboxypenicillins, and Ureidopenicillins)

- ❖ Carboxypenicillins ...carbenicillin and ticarcillin,
 - effective against against **Gram-negative pathogens, including *P aeruginosa***
- ❖ Ureidopenicillin....piperacillin
 - ✓ active against many **Gram-negative bacilli**, such as *Klebsiella pneumoniae* and *P aeruginosa*.

Ureidopenicillin...

- ❖ Combination form**Piperacillin + tazobactam**.
- ❖ Due to the propensity of *P aeruginosa* to develop **resistance** during therapy, an **antipseudomonal β -lactam** is sometimes used in
 - Combined with an **aminoglycoside** or **fluoroquinolone** particularly in infections outside the urinary tract

Adverse Reactions

- ❖ They are **generally well tolerated**, and, unfortunately, this may **encourage inappropriate use**.
- ❖ serious adverse effects are due to **hypersensitivity**
 - ✓ **5–8%** with only a small number of these will have a **serious reaction when given penicillin**.
 - ✓ **< 1%** of who previously received penicillin without incident will have an allergic reaction when given penicillin.
- ❖ Because of the potential for anaphylaxis,
 - ✓ should be **administered with caution**
 - ✓ substitute drug **if history of allergic rxn**.
 - ✓ Penicillin skin testing may also be used to evaluate **Type I hypersensitivity**..

- ❖ In patients with **renal failure**,
 - doses can cause **seizures**.
- ❖ **Nafcillin** is associated with **neutropenia** and interstitial nephritis;
- ❖ **oxacillin** can cause hepatitis; and
- ❖ **methicillin** commonly caused **interstitial nephritis** (and is no longer used for this reason).
- ❖ **Large doses of penicillins** given **orally** may lead to **GI upset, nausea, vomiting, and diarrhea**

❖ **Ampicillin**

- associated with **pseudomembranous colitis**.

❖ **Secondary infections such as vaginal candidiasis may occur.**

❖ **Ampicillin and amoxicillin**

- associated with **skin rashes** when prescribed in the setting of **viral illnesses**, particularly noted during acute **Epstein-Barr virus** infection,

❖ **Piperacillin-tazobactam, vancomycin,**

- greater incidence of **acute kidney injury** compared to **alternate β -lactam agents**

Cephalosporins & cephamycins

❖ Cephalosporins

❖ Similar to penicillins but

- **stable to many bacterial β -lactamases** and,
- **have a broader spectrum of activity.**
- They are not active against *L. monocytogenes*, and
- Only **ceftaroline** has some activity against **enterococci**.

- ❖ Cephalosporins have nucleus of the cephalosporins, **7-aminocephalosporanic acid**
- ❖ the attachment of various R_1 and R_2 d/f potent compounds, many with low toxicity.
- ❖ Cephalosporins have traditionally been **classified into four major groups or generations**, depending **spectrum of antimicrobial activity**

First-generation cephalosporins

- ❖ First-generation cephalosporins
 - **cefazolin, cefadroxil, cephalixin, cephalothin, cephalirin, and cephradine;**
 - These drugs are very active against **Gram-positive cocci**, such as **streptococci and staphylococci**
 - **Anaerobic cocci** (eg, peptococci, peptostreptococci) are usually sensitive,
 - but *Bacteroides fragilis* is not.

- ❖ **Cephalexin** is the oral (of 0.25–0.5 g four times daily) first generation agent widely used in the USA.
- Excretion is mainly by **glomerular filtration** and **tubular secretion into the urine**.
 - Drugs that block tubular secretion, eg, probenecid, may increase serum levels substantially.

Pharmacokinetics & dosage...

- ❖ **Cefazolin** is the only first-generation parenteral.
- ❖ After an intravenous infusion of 1 g
- ❖ Dosage of cefazolin for adults is **0.5–2 g IV** every 8 hours.
- ❖ Cefazolin can also be administered **IM**
 - Excretion is via the **kidney**, and dose adjustments must be made for impaired **renal function**.

Clinical Uses

❖ Oral drugs

–the treatment of **UTI and staphylococcal or streptococcal infections**, including **cellulitis or soft tissue abscess**.

❖ **Cefazolin** penetrates well into most tissues.

❖ A choice for **surgical prophylaxis** and for many **streptococcal** and **staphylococcal** infections requiring intravenous therapy.

❖ **Cefazolin** may be used for infections due to *E coli* or *K pneumoniae*If susceptible.

❖ Cannot be used to treat **meningitis**.

– **Cefazolin** does not penetrate the central nervous system and

❖ It can also be used in patients with **mild penicillin allergy** other than **immediate hypersensitivity**.

Second-generation cephalosporins

Drugs:

- **cefaclor**, cefamandole, cefonicid, **cefuroxime**, **cefprozil**, loracarbef, and ceforanide
- ❖ Structurally related **cephamycins**; **cefoxitin** and **cefotetan**,
 - which have activity against **anaerobes**
- ❖ They are relatively active against organisms inhibited by first-generation drugs, and
 - Extended Gram-negative coverage.**
 - Klebsiella* sp** (including those resistant to first-generation cephalosporins) are usually sensitive

Second-generation cephalosporins...

- ❖ **Cefuroxime and cefaclor**

- active against *H influenzae* but **not against *B fragilis***.

- ❖ In contrast, ***cefoxitin*** and **cefotetan** are active against *B fragilis* and

- ✓ less active against **H influenzae**.

- ❖ Like 1st Gen no member of this group is active against **enterococci** or **P aeruginosa**

- ❖ **cefoxitin** is effective in **extended-spectrum β -lactamases** produced by **E coli** and **Klebsiella sp.**

Pharmacokinetics & Dosage

❖ **Cefuroxime** axetil.

–250–500 mg orally twice daily;

❖ 1-g intravenous infusion,

– IM administration is **painful and should be avoided.**

Clinical Uses

- ❖ Active against
 - **β -lactamase-producing *H influenzae*** or
 - *Moraxella catarrhalis*
- ❖ Used to treat....**sinusitis, otitis, and lower respiratory tract infections.**
- ❖ Active against **anaerobes** (including many *B fragilis* strains),
- ❖ **cefoxitin** and **cefotetan** used
 - to treat mixed anaerobic infections such as **peritonitis, diverticulitis,** and pelvic **inflammatory disease.**

Clinical Uses

❖ Cefuroxime...to treat CAP

—because it is active against **β -lactamase-producing *H influenzae*** and also many **pneumococci**.

❖ Although **cefuroxime** crosses the blood-brain barrier

—it is less effective in treatment of **meningitis** than **ceftriaxone** or **cefotaxime** and should not be used.

Third-generation cephalosporins

- ❖ Drugs.... **cefoperazone, cefotaxime, ceftazidime, ceftizoxime, ceftriaxone, cefixime, cefpodoxime** proxetil, cefdinir, cefditoren pivoxil, ceftibuten, and moxalactam.

Antimicrobial Activity

- ❖ Have expanded **Gram-negative coverage than 2nd G. .**
- ❖ They are also effective against **β -lactamase-producing strains of *Haemophilus* and *Neisseria*.**

Third-generation cephalosporins

- ❖ **Ceftazidime** is the only agent with useful activity against **P.aeruginosa**
- ❖ they are not reliably active against **Enterobacter species**.

Pharmacokinetics & Dosage

- ❖ Third-generation cephalosporins **penetrate body fluids and tissues including CSF**
- ❖ The half-lives of 3rd G. vary
 - ceftriaxone (half-life 7–8 hours)
 - Dose...QD of 15–50 mg/kg/d.
 - **A single daily 1-g dose is sufficient for most serious infections**
 - 2 g BID & QD for **meningitis & endocarditis** respectively

- ❖ Most of them ...(half-life 1–1.7 hours)
- ❖ 2 and 12 g/d (TID-QID) **depending on the severity of infection.**
- ❖ Cefixime 200 mg BID or 400 mg QD **for UTI.**
- ❖ cefixime is no **longer recommended** for the treatment of **uncomplicated gonococcal urethritis and cervicitis.**
 - IM **ceftriaxone + azithromycin** is the regimen of choice for treating most **gonococcal infections**

- ❖ **Ceftriaxone** excretion is mainly through the **biliary tract**, and
 - **no dosage adjustment** is required in renal insufficiency.
- The other third-generation cephalosporins need **dose adjustment**

Clinical Uses

- ❖ used to treat a wide **variety of serious infections** caused by organisms that are **resistant** to most other drugs.
- ❖ **Ceftriaxone** and **cefotaxime** are used for **meningitis**, caused by
 - pneumococci,
 - meningococci, H influenzae,
 - and **susceptible enteric Gram-negative rods**, but not by **L monocytogene**

❖ **Ceftriaxone and cefotaxime...active**

- penicillin-non-susceptible strains of **pneumococci** and
- are recommended for empirical therapy of serious infections that may be caused by these strains
- **Meningitis** caused by strains of pneumococci
 - addition of vancomycin is recommended.
- empirical therapy of sepsis

Drug: **Cefepime** is the **only**

- It is **more resistant to hydrolysis by chromosomal β -lactamase** (eg, those produced by *Enterobacter*)
- But it is hydrolyzed by extended-spectrum β -lactamases...
like third G.

❖ Cefepime has **good** activity against

- P aeruginosa*,
- Enterobacteriaceae,
- methicillin-susceptible *S aureus*, and
- S pneumoniae*

❖ Highly active against

– **Haemophilus** and **Neisseria sp**

Pharmacokinetics

✓ It penetrates well into CSF.

✓ It is cleared by the kidneys and has a half-life of 2 hours,

❖ Dose is 1–2 g infused every BID ;

❖ For complicated infections...2 g TID .

❖ Because of its broad-spectrum activity, cefepime is commonly used empirically for **febrile neutropenia**, in **combination with other agents**.

Adverse effects of cephalosporins

Allergy

- Like penicillins, cephalosporins may elicit a variety of hypersensitivity reactions, including
 - anaphylaxis,
 - fever,
 - skin rashes,
 - nephritis,
 - granulocytopenia, and
 - hemolytic anemia.

Toxicity

- ❖ Local irritation can produce pain after IM injection and **thrombophlebitis** after IV.
- ❖ Renal toxicity,
 - interstitial nephritis and
 - tubular necrosis
- ❖ Cephalosporins that contain a methylthiotetrazole group may cause **hypoprothrombinemia** and bleeding disorders.

Other cell wall synthesis inhibitors

- ✓ **Monobactams**

- [Aztreonam](#)

- ✓ **Carbapenems**

- Doripenem,
 - ertapenem,
 - imipenem, and
 - Meropenem

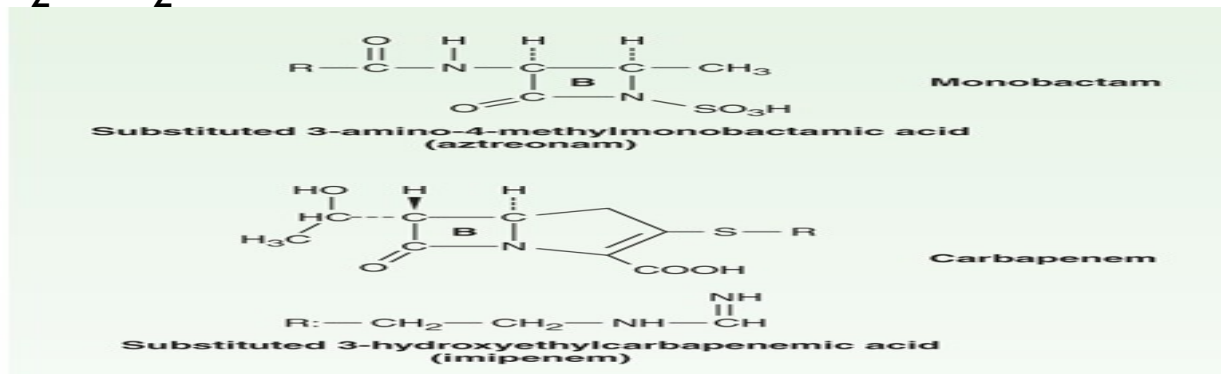
- ❖ **Beta-lactamase inhibitors (clavulanic acid, sulbactam, tazobactam, & avibactam)**

Carbapenems

- Carbapenems are β -lactam antibiotics that contain a fused β -lactam ring and a five-membered ring system that differs from the penicillins in being unsaturated and containing a carbon atom instead of the sulfur atom.
- Has a broader spectrum of activity than do most other β -lactam antibiotics.

Imipenem

- Marketed in combination with ***cilastatin***, a drug that inhibits the degradation of imipenem by a renal tubular dipeptidase
- Derived from a compound produced by *Streptomyces cattleya*. The compound **thienamycin** is unstable, but imipenem is stable.



Mechanism of Action

- Binds to penicillin-binding proteins, disrupts bacterial cell wall synthesis, and causes death of susceptible microorganisms.
- It is very resistant to hydrolysis by most β -lactamases.

Antimicrobial Activity

- Active against
 - *Streptococci* (including pen-resistant), *enterococci* (excluding *E. faecium* and non- β -lactamase-producing pen-resistant strains), *staphylococci* (including penicillinase-producing strains), and *Listeria*
- Most MRSA strains are resistant
- Excellent Activity against the *Enterobacteriaceae*, including organisms that are cephalosporin-resistant
- Most strains of *Pseudomonas* and *Acinetobacter* are inhibited.
- Anaerobes, including *B. fragilis*, are highly susceptible.

➤ Pharmacokinetics

- **Not** absorbed **orally**
- Hydrolyzed rapidly by dipeptidase in **renal** tubule
- Approximately 70% of administered imipenem is recovered in the urine as the active drug

➤ Adverse Reactions

- Nausea and vomiting (most common), Seizures, patients allergic to other β -lactam antibiotics may have hypersensitivity reactions

Therapeutic Uses

- Especially useful for the treatment of infections caused by **cephalosporin-resistant nosocomial bacteria**
- For empirical treatment of **serious infections in hospitalized patients who have recently received other β -lactam**
- Imipenem should **not be** used as monotherapy for infections caused by ***P. aeruginosa*** because of the risk of resistance developing during therapy

➤ Meropenem

- PO inactive
- Not sensitive to renal dipeptidase.
- Similar toxicity to imipenem (less likely to cause seizures)
- G(-) and anaerobes: more active
- G(+): less active
- It is therapeutically equivalence to imipenem.

➤ Ertapenem

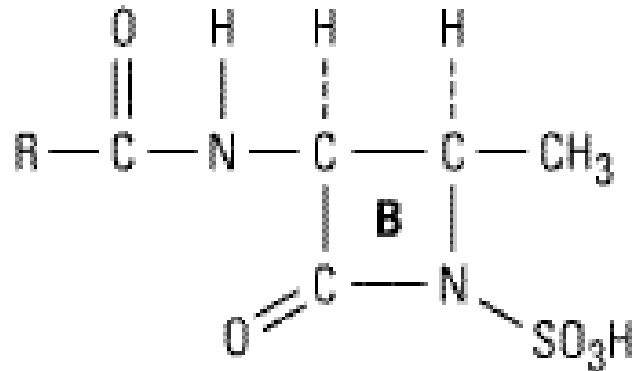
- Have larger serum $t_{1/2}$ that allows once-daily dosing
- Inferior activity **against *Paeruginosa* & *Acinetobacter* spp.**
- Doripenem: more active against ***Paeruginosa***

Monobactams

➤ Aztreonam

- A monocyclic β -lactam compound (a monobactam) isolated from

Chromobacterium violaceum



Mechanism of Action and antimicrobial activity

- Acts in the same way as other β -lactam antibiotics
- Aztreonam is resistant to many of the β -lactamases elaborated by most Gm(-)
- Its antimicrobial activity differs from those of other β -lactams and resembles that of aminoglycosides.
- No activity against Gm(+) bacteria & anaerobic organisms
- Excellent activity against *Enterobacteriaceae* and *P. aeruginosa*.

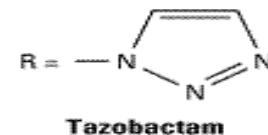
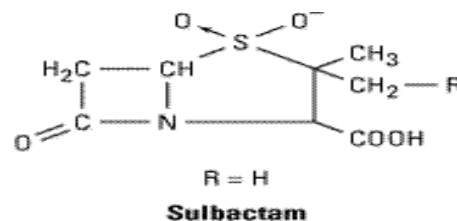
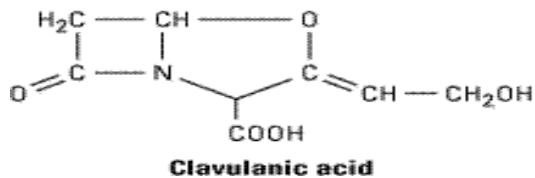
Therapeutic Uses

- Little allergic cross-reactivity with β -lactam antibiotics (possible **exception of ceftazidine**)
- Aztreonam is useful for treating g(-) **infections** that normally would be treated with a β -lactam antibiotic were it not for the history of a prior allergic reaction.

β -Lactamase Inhibitors

- β -Lactamase inhibitors are most active against plasmid-encoded β -lactamases,
- They are inactive at clinically achievable concentrations against β -lactamases induced in Gm(-) bacilli by treatment with 2nd and 3rd GCs
- They resemble β -lactam molecules but have very weak intrinsic antibacterial action.

- A "suicide" inhibitor that **irreversibly binds β -lactamases** produced by a wide range of gram- positive and gram-negative microorganisms
- Used in combination with **β -lactam antibiotics to increase their spectrum of activity.**
 - Amoxicillin + clavulanate (augmentin)
 - Ticarcillin + clavulanate (Timentin)



Vancomycin

- a narrow-spectrum agent active against Gm(+) organisms
- ✓ But lacks of penetration through Gram-negative cell membranes.
- ✓ Not effective against Gm(-) rods, mycobacteria

Mechanisms of Action & Basis of Resistance

- ❖ It inhibits cell wall synthesis by binding to the D-Ala-D-Ala and prevents
 - ✓ further elongation of peptidoglycan and
 - ✓ cross-linking.

The cell membrane is also damaged, which contributes to the antibacterial effect.

Resistance to vancomycin is due to modification of the **D-Ala-D-Ala binding** site of the **peptidoglycan**

Antibacterial Activity

Active against most pathogenic staphylococci, including

✓ β -lactamase producing and those resistant to
nafcillin and **methicillin**

❖ Vancomycin kills staphylococci

Antibacterial Activity...

- ❖ Vancomycin is synergistic with
- ❖ gentamicin and streptomycin.... Against
 - ✓ *Enterococcus faecium* and
 - ✓ *Enterococcus faecalis* strains
- ❖ It is active against many Gram-positive anaerobes including *C difficile*.
- ❖ In combination with gentamicin
 - - alternative regimen for treatment of **enterococcal endocarditis** in a patient with serious **penicillin allergy**

Clinical Uses...

parenteral vancomycin is used in

- ✓ bloodstream infections and **endocarditis** caused by **MRSA** but not effective in **MSSA**
- ✓ **meningitis** caused by a **penicillin-resistant** strain of **pneumococcus**

Oral vancomycin....0.125–0.5 g QID

–to treat colitis caused by C difficile

Adverse drug reactions

- ❖ Most reactions are relatively **minor** and **reversible**.
- ❖ Can cause irritation to the tissue
 - cause phlebitis at the site of injection.
- ❖ Chills and fever may occur.
- ❖ Nephrotoxicity occurs regularly
- ❖ Administration with **ototoxic** or **nephrotoxic** (e.g aminoglycoside,....increases these toxicities.

Adverse drug reactions

- ✓ Vancomycin causes infusion-related ADR.... “red man” syndrome.
- ✓ **prevented by**
 - prolonging the infusion period to 1–2 hours (preferred) or
 - pretreatment with an antihistamine such as **diphenhydramine**.

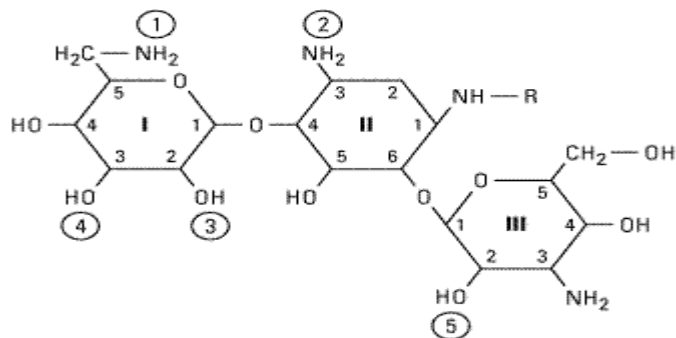
AMINOGLYCOSIDES

Introduction

- Gentamicin, Tobramycin, Amikacin, Netilmicin, Kanamycin, Streptomycin, Neomycin, paromomycin
- Primarily used to treat infections caused by aerobic gram-negative bacteria
- Streptomycin: in the treatment of tuberculosis
- Have bactericidal activity

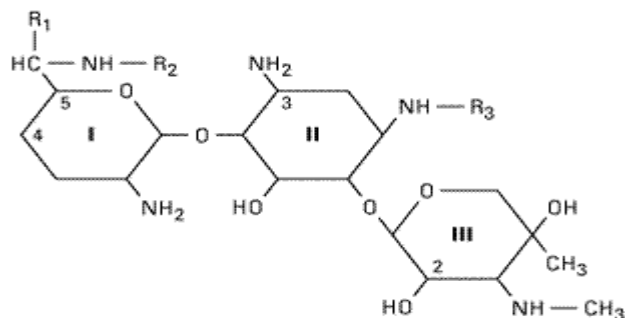
Chemistry

- The aminoglycosides consist of
 - A hexose (aminocyclitol) nucleus [either streptidine (found in streptomycin) or 2-deoxystreptamine (found in all other available aminoglycosides)]
 - Amino sugars attached through glycosidic linkages to the aminocyclitols
- Are water-soluble, stable in solution, and more active at alkaline than at acid pH



Kanamycin R = H

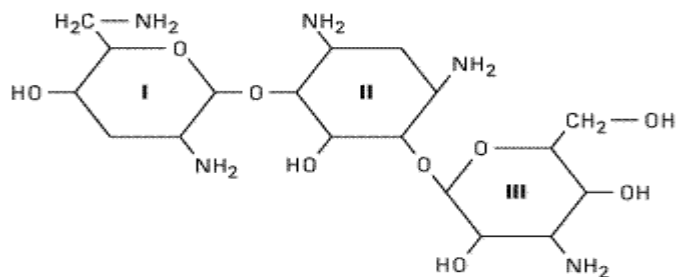
Amikacin R = $\text{C}(=\text{O})-\text{CH}(\text{OH})-\text{CH}_2-\text{CH}_2-\text{NH}_2$



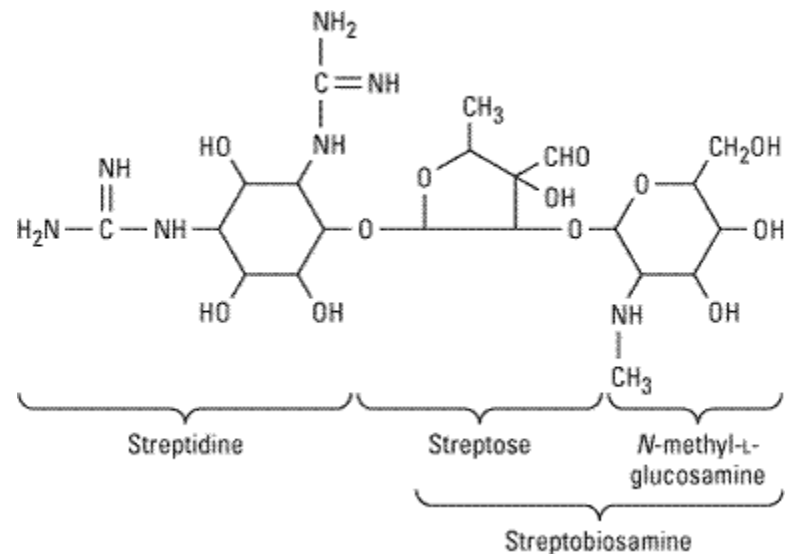
Gentamicin, netilmicin

Gentamicin C₁
Gentamicin C₂
Gentamicin C_{1a}
Netilmicin

Ring I			Ring II
R ₁	R ₂	C4-C5 bond	R ₃
CH ₃	CH ₃	Single	H
CH ₃	H	Single	H
H	H	Single	H
H	H	Double	C ₂ H ₅



Tobramycin



Streptomycin

Mechanism of action

➤ Aminoglycosides

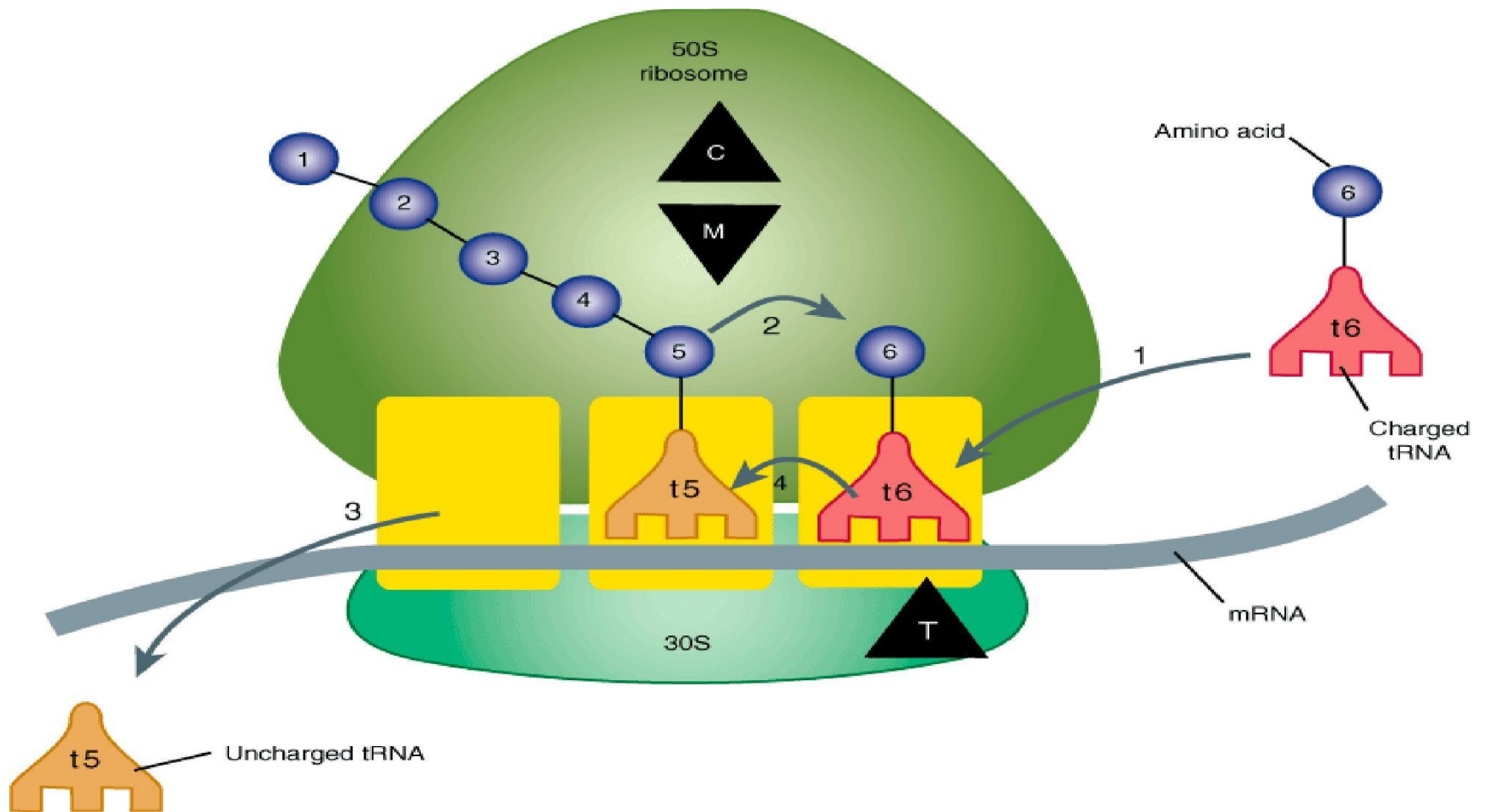
- Are rapidly bactericidal
- Have a **post-antibiotic** effect (their killing action continues when their plasma levels have declined below measurable levels)

➤ Penetration through the outer bacterial membrane

- Outer membrane disruption
- Diffusion through outer membrane porins.

Mechanism of action (cont'd)

- Transport across the **cytoplasmic** membrane depends on **electron transport**
 - Requirement for a membrane electrical potential (interior negative) to drive permeation
 - Antimicrobial activity reduced in the **anaerobic** environment



Mechanism of action (cont'd)

➤ Inside the cell

- Aminoglycosides bind to ribosomes and interfere with protein synthesis
 - ✓ misreading and premature termination of mRNA translation
- Hence, aberrant proteins produced and may be inserted into the cell membrane, leading to altered permeability

➤ Bind to 30S ribosomal subunit and disrupt the normal cycle of ribosomal function by

- blocking formation of **initiation complex**
- **Misreading** of the mRNA template
- **Incorporating** incorrect AA to growing polypeptide chains

Mechanism of Resistance

➤ Resistance could be due to

- Failure to penetrate intracellularly
- Low affinity of the drug for the bacterial ribosome
- Inactivation of the drug by microbial enzymes
 - ✓ Aminoglycoside-modifying enzymes **phosphorylate**, **adenylate**, or **acetylate** specific hydroxyl or amino groups
 - ✓ The metabolites may compete with the parent drug for transport across the inner membrane, but are incapable of binding to ribosomes

Antibacterial Spectrum

- Primarily active against **aerobic Gm(-) bacilli**
- Have little activity against anaerobic microorganisms or facultative bacteria under anaerobic conditions.
- Limited action against most gram-positive bacteria
 - Produce **synergistic bactericidal effect *in vitro*** (against enterococci, streptococci, and staphylococci) when **combined with a cell wall-active agent**

Pharmacokinetics

➤ Absorption

- Very poorly absorbed from the GIT.
- Absorbed rapidly from IM sites of injection.
 - ✓ In critically ill patients, especially those in shock, absorption of drug may be reduced from intramuscular sites because of poor perfusion

➤ Distribution

- Do not penetrate into most cells and has negligible plasma albumin binding (Except for streptomycin)
- $V_d \cong$ the volume of extracellular fluid
- High concentrations are found in **renal cortex and inner ear** (likely contributor to the nephrotoxicity and ototoxicity)
- **Streptomycin** and **tobramycin** can cause **hearing** loss in children born to women who receive the drug during pregnancy.
 - ✓ Recommendation: use with caution during pregnancy and only for strong clinical indications in the absence of suitable alternatives.

➤ Elimination.

- The aminoglycosides are excreted almost entirely by glomerular filtration

➤ Dosing

- Total daily dose as a **single injection** or two or three equally divided doses
- **Total daily dose as a single injection is associated with less toxicity and is just as effective as multiple- dose regimens**
 - ✓ More drug accumulates with higher plasma concentrations, particularly at trough, and with prolonged periods of exposure.

Streptomycin

- Is generally **less active than other members of the class**
- Administered by **deep IM or IV**
- Dose adjustment in **renal impairment**
- Therapeutic Uses
 - ***Bacterial Endocarditis***
 - ✓ Streptomycin and penicillin in combination
 - ***Tuberculosis***

Gentamicin

- First choice due to low cost and reliable activity against **all but the most resistant g(-) aerobes.**
- Available for parenteral, ophthalmic, and topical
- Therapeutic Uses (Gentamicin, tobramycin, amikacin, and netilmicin)
 - *Urinary Tract Infections*
 - *Pneumonia*
 - *Meningitis*
 - *Bacterial Endocarditis*
 - *Sepsis*

Tobramycin

➤ Therapeutic Uses

- Same as those for gentamicin
- The preferred aminoglycoside for treatment of serious infections caused by *P. aeruginosa* (usually in combination with an antipseudomonal β -lactam antibiotic)
- Poor activity in combination with penicillin against many strains of enterococci.
- Ineffective against mycobacteria

Amikacin

- **Broadest spectrum of activity in the group**
- **Resistant to many of the aminoglycoside-inactivating enzymes**
- **Therapeutic Uses**
 - The preferred agent for the **initial treatment of serious nosocomial Gm(-) bacillary infections in hospitals**
 - Active against the majority of **aerobic g(-) bacilli** in the **community and the hospital**
 - Less active than **gentamicin against enterococci**
 - It is active against ***M. Tuberculosis***

Untoward effects

➤ **Vestibular & auditory ototoxicity**

- Vestibular and auditory dysfunction can follow the administration of any of the aminoglycosides

➤ **Nephrotoxicity**

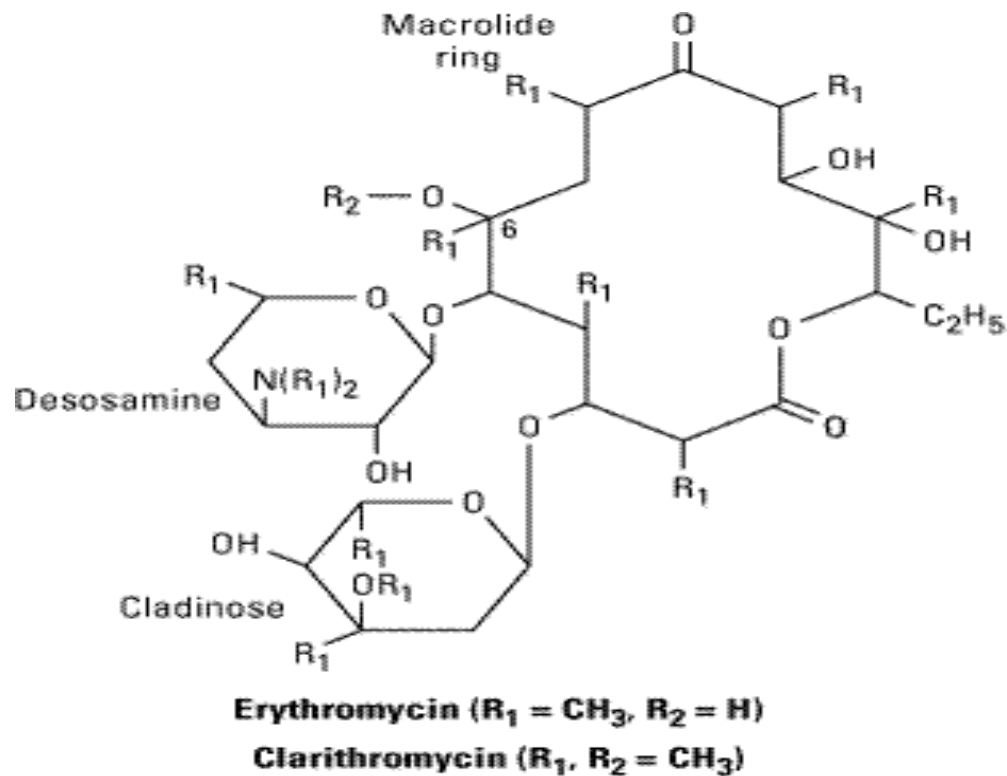
➤ **Neuromuscular Blockade**

- And the associated apnea
- Decreasing potency for blockade: **neomycin**, kanamycin, amikacin, gentamicin, and tobramycin.

MACROLIDES

Introduction

- *Contain a large 14 or 15 membered lactone ring to which one or more deoxy sugars are attached.*
- Include *erythromycin, clarithromycin, azithromycin, and oleandomycin*
- **Azithromycin**
 - Improve acid stability and tissue penetration and broaden the spectrum of activity



- **Clarithromycin** differs from erythromycin only by methylation of the hydroxyl group at the 6 position,
- Azithromycin differs by the addition of a methyl-substituted nitrogen atom into the lactone ring.

Mechanism of Action

- Inhibit protein synthesis by binding reversibly to **50S** ribosomal subunit
- Inhibits the **translocation** step where in a newly synthesized peptidyl tRNA molecule moves from the **acceptor** site on the ribosome to the **peptidyl donor site**.
- **Increased** antimicrobial activity at **alkaline pH** (more permeability of the un-ionized form)

Antibacterial Activity

- Erythromycin usually is **bacteriostatic**, but may be **bactericidal** in high concentrations
- The antibiotic is most active *in vitro* against aerobic **gram-positive cocci and bacilli**.
- **Cross-resistance** among macrolides is complete
- **Staphylococci** are not reliably sensitive to erythromycin.

Antibacterial Activity (cont'd)

- Erythromycin is **inactive** against most **aerobic enteric Gm-negative bacilli**.
- Macrolides are active against *Campylobacter jejuni*.
- Erythromycin is active against *M. Pneumoniae*

Antibacterial Activity (cont'd)

- Clarithromycin is more potent than erythromycin against
 - sensitive strains of streptococci and staphylococci,
 - has modest activity against *H. influenzae* and *N. gonorrhoeae*.
- Azithromycin:
 - less active than erythromycin against Gm(-) organisms and
 - slightly more active against *H. influenzae* and *Campylobacter* spp.
- Azithromycin & clarithromycin have enhanced activity against
 - *M. avium-intracellulare*
 - Some protozoa (e.g., *T gondii* and *Plasmodium* spp.)

Mechanism of resistance

- Resistance to macrolides usually results from
 1. Drug efflux by an active pump mechanism
 2. Ribosomal protection by inducible or constitutive production of **methylase** enzymes, which modify the ribosomal target and decrease drug binding
 3. Macrolide hydrolysis by **esterases** produced by Enterobacteriaceae
 4. Chromosomal mutations that alter a 50S ribosomal protein

Pharmacokinetics

- Absorbed from the intestinal tract,
 - Presence of food interferes with absorption (Azithromycine)
 - Erythromycin is relatively more acid labile
 - Serum levels of clarithromycin and azithromycin are low, but they concentrate in tissue and reach high levels.
 - Erythromycin and azithromycin are excreted primarily in bile
 - **Clarithromycin** is metabolized to the biologically active 14-OH metabolite and is eliminated largely by the kidney.
- azithromycin (68 hours),
 $t_{1/2}$: erythromycin (1.4 hours), clarithromycin (3 to 7),

Clinical uses

- *Mycoplasma pneumonia* infections
 - A macrolide or TTC is the drug of choice
- *Chlamydial* Infections
 - Can be treated effectively with any of the macrolides
- Diphtheria (*Corynebacterium diphtheriae*)
 - erythromycin
- Pertussis
 - Drug of choice for treatment & prophylaxis

Clinical uses (cont'd)

➤ Streptococcal infections

- Alternatives in patients who are allergic to penicillin. **Penicillin-resistant strains of *S. Pneumoniae*** are very likely to be resistant to macrolides

➤ Staphylococcal infections

➤ Campylobacter infections

- largely replaced by Fluoroquinolones in adults
- Erythromycin remains useful for treatment of *Campylobacter gastroenteritis* in children

Clinical uses (cont'd)

➤ ***Tetanus***

- Erythromycin may be given to eradicate ***Clostridium tetani*** in patients allergic to penicillin

➤ ***Mycobacterial Infections***

- Clarithromycin or azithromycin is recommended as first- line therapy for **prophylaxis and treatment of disseminated infection caused by *M. avium-intracellulare*** in AIDS patients and for treatment of pulmonary disease in non-HIV-infected patients

Adverse Effects

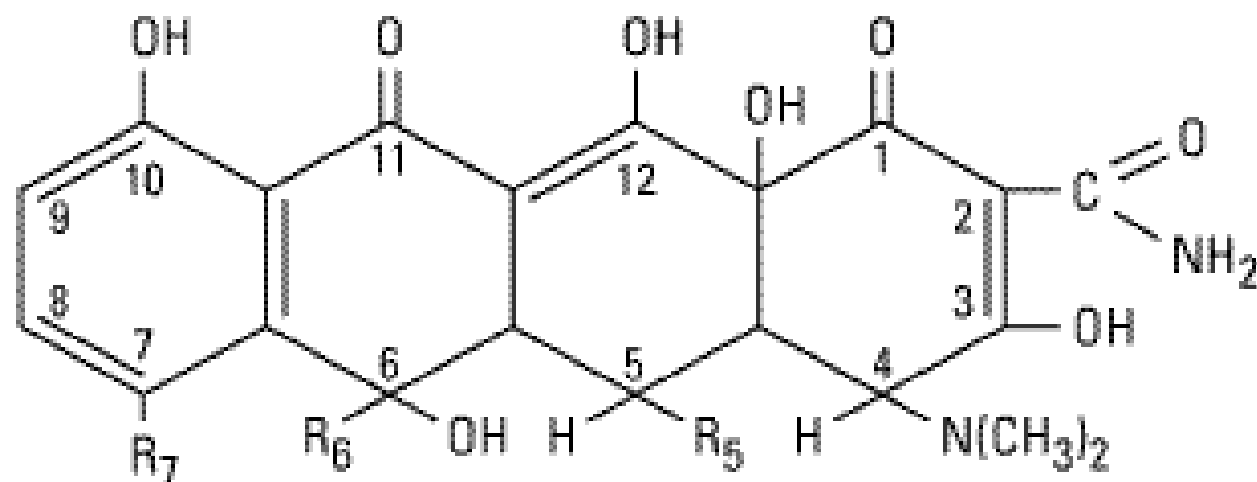
- The incidence of side effects associated with erythromycin therapy is very low.
 - Mild GI upset with nausea, diarrhea, and abdominal pain (more common)
 - Rashes are seen infrequently
 - Thrombophlebitis may follow IV administration
 - Transient impairment of hearing.
- **Cholestatic hepatitis** (characterized by fever, enlarged and tender liver, hyperbilirubinemia, dark urine, eosinophilia, elevated serum bilirubin, and elevated transaminase levels)
- The drugs **inhibits hepatic microsomal enzymes** and interfere with the actions of various drugs

TETRACYCLINES

(TTC)

Introduction

- Produced by different species of *Streptomyces*
 - *OxyTTC* is a natural product elaborated by *Streptomyces rimosus*
 - TTC is a semisynthetic derivative of chlorTTC
 - *Demeclocycline* is the product of a mutant strain of *Strep. Aureofaciens*
 - *Methacycline*, *doxycycline*, and *minocycline* are semisynthetic derivatives.
- *Chlortetracycline*, the prototype of the class, was introduced in 1948

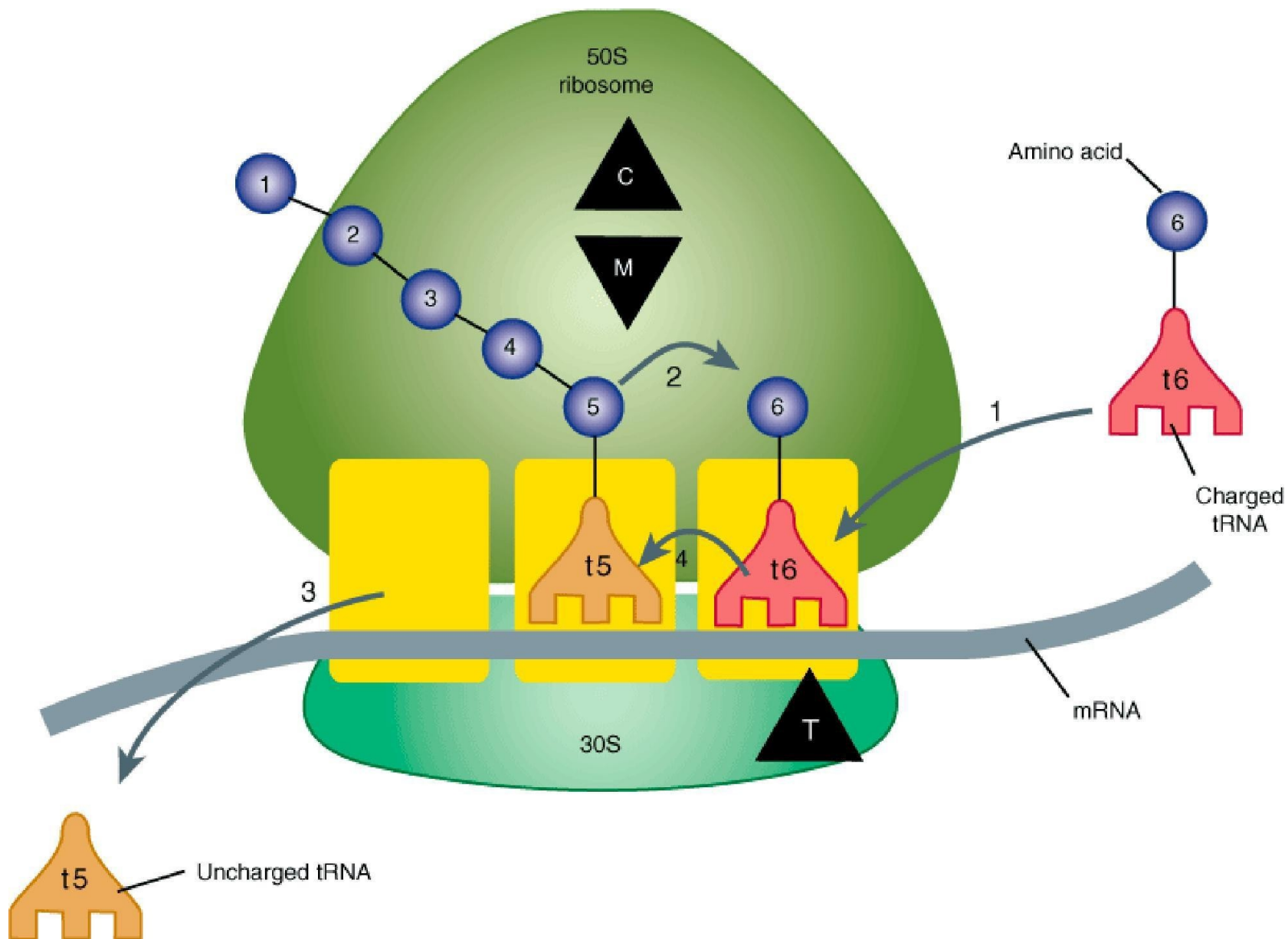


	R₇	R₆	R₅	Renal Clearance (mL/min)
Chlortetracycline	— Cl	— CH ₃	— H	35
Oxytetracycline	— H	— CH ₃	— OH	90
Tetracycline	— H	— CH ₃	— H	65
Demeclocycline	— Cl	— H	— H	35
Methacycline	— H	≡ CH ₂ *	— OH	31
Doxycycline	— H	— CH ₃ *	— OH	16
Minocycline	— N(CH ₃) ₂	— H	— H	10

*There is no — OH at position 6 on methacycline and doxycycline.

Mechanism of Action

- **TTCs enter Gm(-) bacteria by passive diffusion through the porin channels of the outer cell membrane**
-
- **By active transport *via* an energy-dependent system that pumps all TTCs across the cytoplasmic membrane**
- **TTCs inhibit bacterial protein synthesis by binding to the 30S bacterial ribosome and
 - **preventing access of aminoacyl tRNA to the acceptor (A) site on the mRNA-ribosome complex****
- **Entry into Gm(+) bacteria requires metabolic energy, but is not as well understood**



Effects on Pathogenic Microorganisms

- Are **bacteriostatic antibiotics** active against a wide range of **aerobic** and **anaerobic** Gm(+) & Gm(-) bacteria
- Are also effective against ***Rickettsia***, ***Mycoplasma pneumoniae***, ***Chlamydia***, some atypical mycobacteria, and
- ***Plasmodium* spp.**, that are resistant to cell-wall-active antimicrobial agents

Effects on Intestinal Flora

- Many of the TTCs are incompletely absorbed from the GIT, and high concentrations in the bowel can alter enteric flora.
- Sensitive aerobic & anaerobic microorganisms & Gm(+) spore-forming bacteria are markedly suppressed
- Overgrowth of TTC-resistant microorganisms
Pseudomonas spp.), enterococci, *Proteus*, & *Candida* spp.)

Resistance to the Tetracyclines

➤ The main resistance mechanisms

- 1. Decreased accumulation of tetracycline as a result of either decreased antibiotic influx or acquisition of an energy-dependent efflux pathway;**
- 2. Production of a ribosomal protection protein that displaces tetracycline from its target**
- 3. Enzymatic inactivation of tetracyclines**

Pharmacokinetics

➤ Absorption

- Partially absorbed from the stomach and upper GIT
- Impaired by Food (*all except doxycycline and minocycline*)
- *TTCs form insoluble chelates with calcium, magnesium, and other metal ions* ⇒ interfere with absorption

➤ Distribution

- Distributed throughout body tissues and fluids
 - ✓ Minocycline and doxycycline (highest lipid solubility), oxyTTC (the least)

➤ Elimination

- TTCs are excreted mainly in **bile and urine**
- Ten to 50 % of various TTCs is excreted into the urine.
- **Doxycycline** is eliminated by **nonrenal mechanisms** ⇒ **TTC of choice** in renal insufficiency.

Classification

➤ Based on serum half-lives, TTCs are Classified

- Short-acting
 - ✓ Serum half-life of 6-8 hours
 - ✓ ChlorTTC (30% PO Abs), TTC, oxyTTC (60-70%)
- Intermediate-acting)
 - ✓ Serum half-life of 12 hours
 - ✓ Demeclocycline and methacycline (60-70%)
- long-acting
 - ✓ Serum half-life of 16-18 hours
 - ✓ Doxycycline and minocycline (995-100%)
 - ✓ Almost complete absorption and slow excretion of these drugs allow for once daily dosing.

Therapeutic Uses

➤ ***Rickettsial Infections***

- Doxycycline is the drug of choice

➤ **Mycoplasma Infections**

➤ ***Chlamydial infections***

➤ ***Trachoma***

➤ ***Sexually Transmitted Diseases***

- Because of resistance, doxycycline no longer is recommended for gonococcal infections

➤ ***Anthrax (bacillus anthracis)***

- Doxycycline: for prevention or treatment of anthrax
- *Propionibacterium acnes*

Therapeutic Uses (cont'd)

- Cholera

- ✓ Doxycycline

- ***Other Bacillary Infections***

- Ineffective in infections caused by *Shigella*, *Salmonella*, or other Enterobacteriaceae

Therapeutic Uses (cont'd)

➤ ***Coccal Infections***

- Community strains of MRSA often are susceptible to TTC, doxycycline, or minocycline
- Doxycycline effective for empirical therapy of community-acquired pneumonia

➤ Are sometimes employed in the treatment of protozoal infections (*Entamoeba histolytica*, *Plasmodium falciparum*)

Adverse effects

➤ Gastrointestinal Adverse Effects

- Nausea, vomiting, and diarrhea
- modification the normal flora with overgrowth of pseudomonas, proteus, staphylococci, resistant coliforms, clostridia, and candida

➤ Bony Structures and Teeth

- **Deposition in calcium deposited in newly formed bone or teeth**

Adverse effects (cont'd)

➤ Liver Toxicity

- Especially during pregnancy

➤ Kidney Toxicity

- Aggravate azotemia (BUN,Cr) in patients with renal disease

➤ Photosensitization

➤ **Contraindications**

- In pregnant women
- Children under the age of 8 years

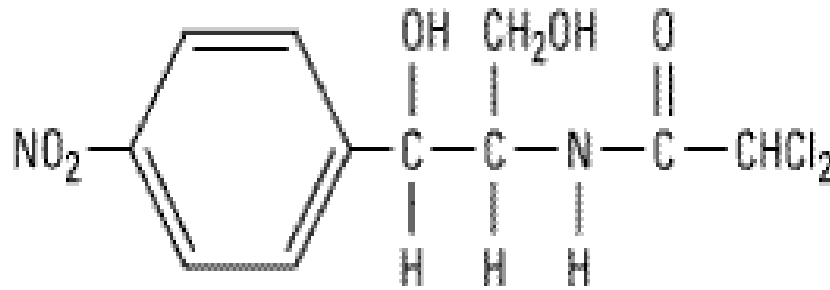
➤ **Therapeutic selection**

- Little difference in clinical response among the various TTCs.

CHLORAMPHENICOL

Introduction

- Crystalline chloramphenicol is a neutral, stable compound

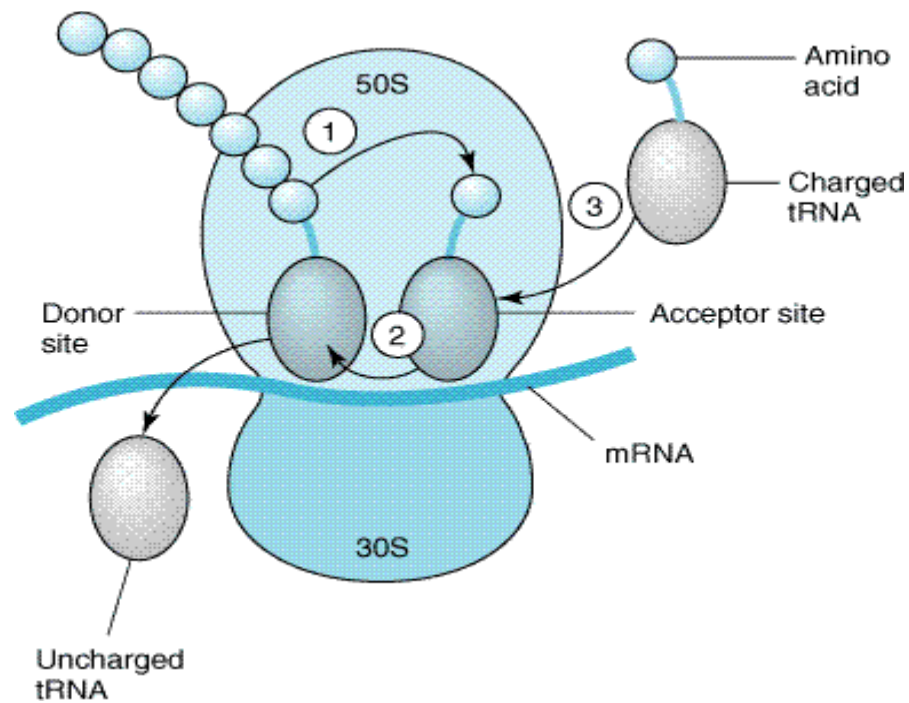


Chloramphenicol

- Contains a **nitrobenzene** moiety and is a derivative of **dichloroacetic acid**.
- The biologically active form is **levorotatory**.

Mechanism of Action

- It inhibits protein synthesis in bacteria, and to a lesser extent, in eukaryotic cells.
 - Binds reversibly to the 50S ribosomal subunit
 - the drug **apparently prevents the binding of the amino acid- containing end of the aminoacyl tRNA to the acceptor site on the 50S ribosomal subunit.**
 - The interaction between **peptidyltransferase** and its amino acid substrate cannot occur, and **peptide bond formation is inhibited**



- Steps in bacterial protein synthesis and targets of (1) chloramphenicol; (2) macrolides, clindamycin, and (3) tetracyclines

Pharmacokinetics

- Crystalline CAF is rapidly and completely absorbed After oral administration
- **widely distributed to virtually all tissues and body fluids, including the CNS and CSF s**
- It penetrates cell membranes readily
- Most of the drug is inactivated either by conjugation with glucuronic acid or by reduction to inactive aryl amines

Resistance to Chloramphenicol

- Resistance is caused due to
 - Plasmid-encoded **acetyltransferase** that inactivates the drug (Acetylated derivatives fail to bind to bacterial ribosomes)
 - **Decreased permeability of the drug**
 - **Ribosomal mutation.**

Antimicrobial Actions

- Broad spectrum antimicrobial activity
- Effective against **Gm(+)** and **Gm(-)** bacteria, including *Rickettsia*, *Mycoplasma*, and *Chlamydia* spp.
- Also effective against most **anaerobic** bacteria, including *Bacteroides fragilis*.

Therapeutic Uses

➤ Typhoid Fever

- ✓ *3rd GC and quinolones are drugs of choice*

➤ Bacterial meningitis

- ✓ 3rd GC have replaced CAF
- ✓ An alternative in patients allergic to β -lactams and in developing countries

➤ Rickettsial Diseases

- ✓ The CAF is a preferred agent
 - In patients **allergic** to TTCs,
 - in **reduced renal function**,
 - in **pregnant women**, and
 - in **children younger than 8** years of age who require prolonged or repeated courses of TTCs therapy, **CAF is the drug of choice**

Untoward effects

- **CAF inhibits the synthesis of proteins of the inner mitochondrial membrane, probably by inhibiting the ribosomal peptidyltransferase.**

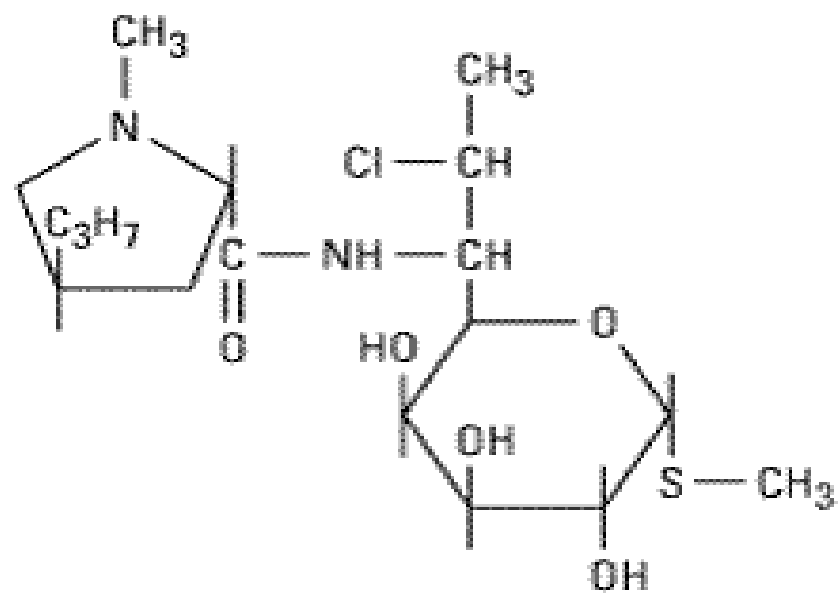
Untoward effects (cont'd)

- Newborn infants cannot adequately **conjugate CAF** to form the **glucuronide**
 - ⇒ High levels of free CAF cause the *gray baby syndrome*.
 - Abdominal distention, vomiting, progressive cyanosis, irregular respiration, hypothermia, and vasomotor collapse
- Effects on hematopoietic system: manifested by **bone marrow depression** or idiosyncratic aplastic anemia.
 - Bone marrow depression is dose related and is characterized by **anemia**, **leukopenia** or **thrombocytopenia**, but it is reversible on discontinuation of CAF.
 - Aplastic anemia is usually fatal

Drug Interactions

- CAF inhibits hepatic CYP450s and prolongs the $t_{1/2}$ of drugs metabolized by this system (*warfarin*, *dicumarol*, *phenytoin*, *chlorpropamide*, antiretroviral protease inhibitors, *rifabutin*, and *tolbutamide*)
- *Phenobarbital*, *rifampin* potently induce CYPs and shorten the $t_{1/2}$ of CAF

Clindamycin



Clindamycin

Mechanism of Action

- The binding site for clindamycin on the **50S** subunit of the bacterial ribosome is identical with that for erythromycin.
- Clindamycin inhibits protein synthesis by interfering with the formation of **initiation complexes** and with **aminoacyl translocation reactions**.

Antibacterial Activity

- Streptococci, staphylococci, and pneumococci are inhibited by clindamycin
- Enterococci and Gm(-) aerobic organisms are resistant (in contrast to their susceptibility to erythromycin) because of poor permeability of the outer membrane.
- Bacteroides species and other anaerobes, both Gm(+) and Gm(-), are usually susceptible.
- *Clostridium difficile* is resistant.

Pharmacokinetics

➤ Absorption

- Nearly completely absorbed following oral administration
- Food in the stomach does not reduce absorption significantly

➤ Distribution

- Widely distributed in many fluids and tissues
- Readily crosses the placental barrier

➤ Excretion

- 10% is excreted unaltered in the urine, and small quantities are found in the feces.
- Inactivated by metabolism to **N-demethylclindamycin** and **clindamycin sulfoxide**, which are excreted in the urine and bile

Therapeutic Uses

- **Treatment of severe anaerobic infection** caused by bacteroides and other anaerobes
- In combination with aminoglycoside or cephalosporin,
 - To treat **penetrating wounds** of the abdomen and the gut
- **Clindamycin plus primaquine** is an effective alternative to co-trimoxazole for moderate to moderately severe ***Pneumocystis carinii pneumonia* in AIDS patients.**
- Also used in combination with pyrimethamine for AIDS- related **toxoplasmosis** of the brain.

Untoward Effects

- **Diarrhea** (in 2% to 20% subjects)
- **Pseudomembranous colitis** caused by the toxin from *C. difficile*
 - Characterized by abdominal pain, diarrhea, fever, and mucus and blood in the stools.
- Skin rashes, **Stevens-Johnson syndrome**, impaired liver function, granulocytopenia, thrombocytopenia, and anaphylactic reactions
- **Local thrombophlebitis** with IV administration
- Can inhibit neuromuscular transmission and may potentiate the effect of a **neuromuscular blocking**

LINEZOLID

- Linezolid is a synthetic oxazolidinone

Mechanism of action

- **Linezolid binds to the bacterial 23S ribosomal RNA of the 50S subunit, thereby inhibiting the formation of the 70S initiation complex.**
- **developed to combat resistant gram-positive organisms, such as**
 - **MRSA, VRE, and penicillin-resistant streptococci.**
 - **Corynebacterium species**
 - **Listeria monocytogenes**
 - **Clostridium perfringens**
 - **Mycobacterium tuberculosis**

- linezolid is **bacteriostatic**.
- It is bactericidal against **streptococci**.
- Linezolid is an alternative to **daptomycin** for infections caused by VRE.
- Use of linezolid for the treatment of MRSA bacteremia is not recommended.

Pharmacokinetics

- Linezolid is completely **absorbed after oral** administration.
- An IV preparation is also available.
- The drug is widely distributed throughout the body.
- It is known that it is metabolized via oxidation to two inactive metabolites.
- The drug is excreted both by renal and nonrenal routes.
- No dose adjustments are required for renal or hepatic dysfunction.

Adverse effects

- Gastrointestinal upset, nausea, diarrhea, headache, and rash.
- Thrombocytopenia: in patients taking the drug for longer than 10 days.
- Serotonin syndrome
- Irreversible **peripheral neuropathies and optic neuritis** (causing blindness) have been associated with greater than 28 days of use

Self-test questions

1. Which of the following antibiotic combinations is inappropriate based on antagonism at the same site of action?

- A. Clindamycin and erythromycin.
- B. Doxycycline and amoxicillin.
- C. Tigecycline and azithromycin.
- D. Cipronoxacin and amoxicillin.

2 Children younger than 8 years of age should not receive tetracyclines because these agents:

- A. Cause rupture of tendons.
- B. Deposit in tissues undergoing calcification.
- C. Do not cross into the cerebrospinal fluid.
- D. Can cause aplastic anemia.

3 A 30-year-old pregnant female has cellulitis caused by MRSA. Which of the following antibiotics would be the most appropriate option for outpatient therapy?

- A. Doxycycline.
- B. Clindamycin.
- C. Erythromycin
- D. Tigecycline.

4 A patient is being discharged from the hospital on a 3-week course of clindamycin. Which of the following potential adverse effects should be discussed with her?

- A. Hyperbilirubinemia.
- B. Nephrotoxicity.
- C. Clostridium difficile diarrhea

SULFONAMIDES

Chemistry

- Has the basic **sulfanilamide** nucleus (which is structurally similar to **p-aminobenzoic acid**)
 - Derivatization: attaching substituents to the **amido** group ($-\text{SO}_2\text{-NH-R}$) or the **amino** group ($-\text{NH}_2$)
- Sulfonamides tend to be much more soluble at alkaline than at acidic pH.

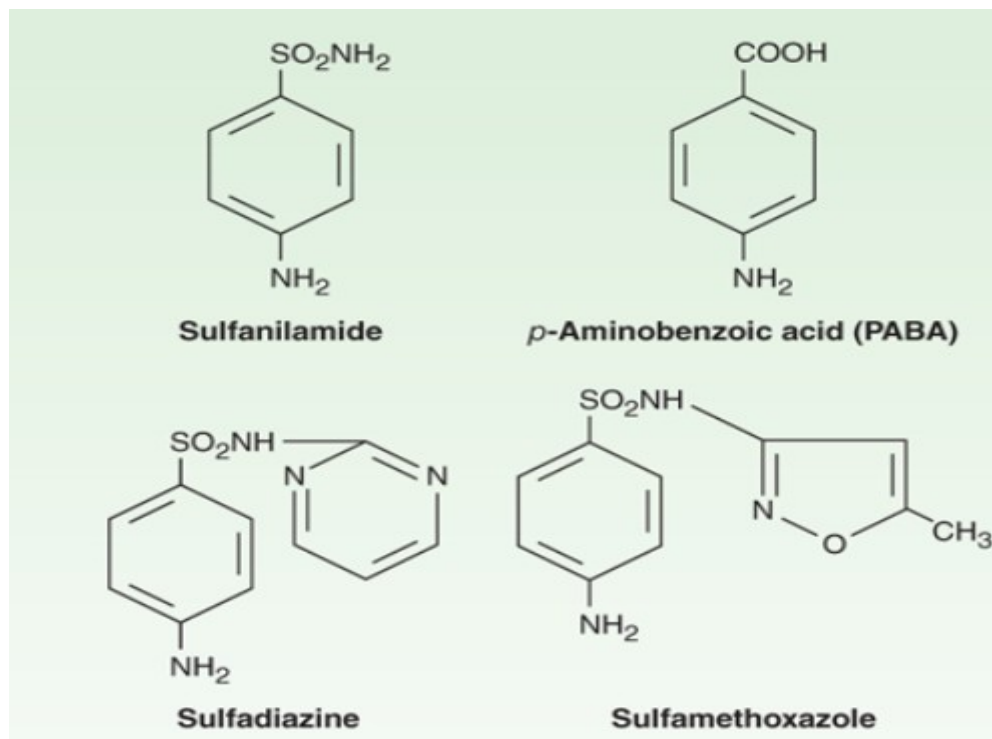


Figure: Structures of some sulfonamides and *p*-aminobenzoic acid

Mechanism of Action & Antimicrobial Activity

- ❖ Sulfonamide-susceptible organisms, cannot use exogenous folate
 - must synthesize it from PABA.
- ❖ sulfonamides inhibit **dihydropteroate synthase** and **folate production**.

Spectrum of activity

- ❖ **Rickettsiae** are not inhibited by sulfonamides

Gram (+)	Gram (-)	
Staphylococcus	E.coli	Nocardia sp
	K. pneumoniae	Chlamydia trachomatis
	Salmonella,	
	Shigella,	
	Enterobacter sp	

Mechanism of Action & Antimicrobial Activity

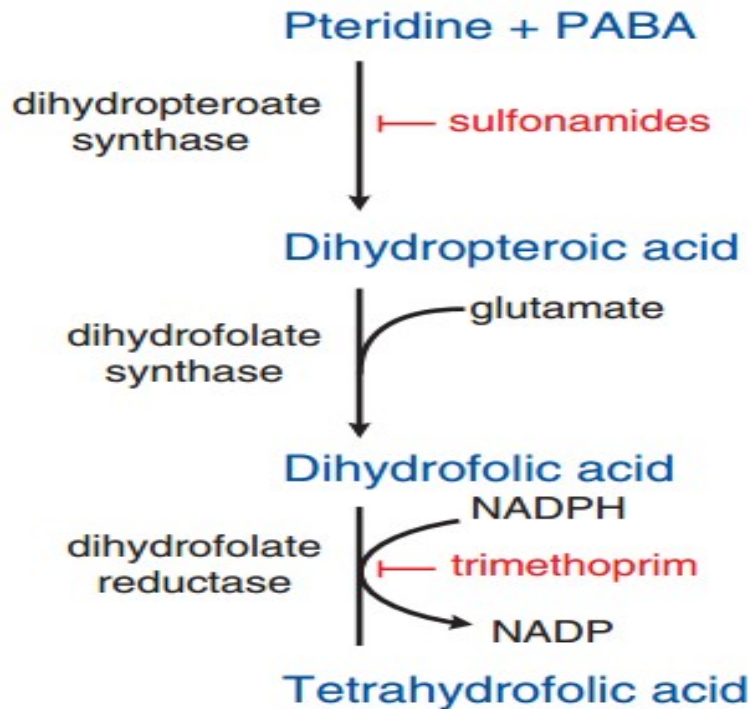


Figure: Actions of sulfonamides and trimethoprim.

- ❖ sulfonamide with an inhibitor of dihydrofolate reductase (trimethoprim or pyrimethamine)
 - provides synergistic activity because of sequential inhibition of folate synthesis

Resistance

- ❖ **Some bacteria lack the enzymes required for folate synthesis from PABA and depend on exogenous sources of folate;**
 - not susceptible to sulfonamides.
- ❖ Sulfonamide resistance may also occur as a result of
 - cause overproduction of PABA,
 - cause production of a folic acid-synthesizing enzyme that has low affinity for sulfonamides
 - impair permeability to the sulfonamide..

Pharmacokinetics

- ❖ Sulfonamides can be divided into three major groups:
 - oral, absorbable;
 - oral, non-absorbable; and
 - topical.

Oral absorbable sulfonamides

- ✓ absorbed from the stomach and small intestine
- ✓ Protein binding20% to over 90%.

Metabolism... acetylated or glucuronidated in the liver.

- ✓ Excretion,,,,, excreted in the urine, mainly
 - must be reduced in patients with significant renal failure.

Clinical Uses

- ❖ Sulfonamides are infrequently used as single agents.
- ❖ The combination of trimethoprim-sulfamethoxazole is the drug of choice for
 - Pneumocystis jiroveci (formerly P carinii) pneumonia, toxoplasmosis, and
 - nocardiosis.

Oral absorbable

Sulfamethoxazole and Sulfisoxazole is a commonly

- For treatment of UTI

Sulfadiazine + pyrimethamine

- first-line therapy for treatment of acute toxoplasmosis.

Sulfadoxine + pyrimethamine (Fansidar)

- used as a second-line treatment for malaria

.

Oral nonabsorbable agents

Sulfasalazine (salicylazosulfapyridine): sulfapyridine and 5-aminosalicylate

- is widely used in ulcerative colitis,
 - enteritis, and
 - other inflammatory bowel disease

Topical Agents

Sodium sulfacetamide ophthalmic solution

- For treatment of bacterial conjunctivitis and
- as adjunctive therapy for trachoma.

Silver sulfadiazine is a less toxic

- preferred for prevention of infection of burn wounds.

Adverse Reactions

The most common adverse effects are

- ✓ fever, skin rashes, exfoliative dermatitis, photosensitivity, urticaria, nausea, vomiting, diarrhea,

Stevens-Johnson syndrom... uncommon (<1%) but
—serious and potentially fatal.

❖ Other unwanted effects include

- ✓ stomatitis, conjunctivitis, arthritis, hematopoietic disturbances), hepatitis, and, rarely, polyarteritis nodosa and psychosis.

Adverse Reactions...

Urinary Tract Disturbances

Sulfonamides at neutral or acid urine pH ,

- producing crystalluria, hematuria, or obstruction.

- But rare with the more soluble sulfonamides (eg, sulfisoxazole).

- Sulfadiazine and sulfamethoxazole ..relatively insoluble in (low pH) cause....crystalluria,

- ❖Crystalluria is treated by administration of sodium bicarbonate and fluids

Adverse Reactions...

Hematopoietic Disturbances

- ❖ Sulfonamides can cause
 - hemolytic or aplastic anemia,
 - granulocytopenia,
 - thrombocytopenia, or
 - leukemoid reactions.
- ❖ Sulfonamides may provoke **hemolytic reactions in G-6-P dehydrogenase deficiency.**
- ❖ Sulfonamides increase the **risk of kernicterus in newborns.**

Trimethoprim & trimethoprim-sulfamethoxazole combination

Trimethoprim is a much less **efficient inhibitor of mammalian dihydrofolic acid reductase**.

- ✓ combination of trimethoprim and sulfamethoxazole is often bactericidal,
- ✓ But sulfonamide alone is **bacteriostatic**

.

Pharmacokinetics

- ❖ Trimethoprim given orally/IV, alone or in combination with **sulfamethoxazole**
- ✓ Both have a similar half-life.
- ✓ Given as (1:5 ration) TMP-SMX
- ✓ 1 part of trimethoprim is given with 5 parts of sulfamethoxazole
- ✓ 30–50% of the sulfonamide and 50–60% TMP of the ..excreted in the urine **within 24 hours**.
 - Dose be reduced by 50 % with creatinine **clearances of 15–30 mL/min**.
- ❖ **Trimethoprim** concentrates in **prostatic fluid and in vaginal fluid**,
 - has more antibacterial activity in this areas.

Clinical uses

Oral Trimethoprim

✓ Trimethoprim 100 mg twice daily in acute UTI

❖ Oral Trimethoprim-Sulfamethoxazole (TMP-SMZ)
effective treatment for a wide variety of infections

- PCP ——— Respiratory infections
- UTI ——— uncomplicated skin and soft tissue
- Prostatitis ——— nontuberculous mycobacteria.
- Shigella
- Salmonella
- MSSA and MRSA
- ❖ In UTI.....UP to 30% in E.coli pneumococci
 - ✓ consider to resistance before using as an empiric

Clinical uses...

❖ Infections with *P jiroveci* and *Nocardia*

✓ Treated with high doses of either the PO or IV (15–20 mg/kg/d).

Intravenous Trimethoprim-Sulfamethoxazole

❖ *for moderately severe to severe pneumocystis pneumonia*

❖ for Gram-negative bacterial sepsis

– replaced by extended spectrum β -lactams and fluoroquinolones.

❖ Used as an alternative for multidrug-resistant infection

❖ serious *Listeria* infections (unable to tolerate ampicillin).

–10–20 mg/kg/d of the trimethoprim component.

Oral Pyrimethamine with Sulfonamide

❖ They are used in the treatment of **toxoplasmosis**.

Dosage.....sulfadiazine is + **pyrimethamine**

✓ Leucovorin (folinic acid)....10 mg orally each day

—to minimize bone marrow suppression

.

Adverse Effects

Trimethoprim

- ❖ megaloblastic anemia,
- ❖ leukopenia, and
- ❖ Granulocytopenia
- ❖ Nausea and vomiting, drug fever, vasculitis, renal damage,
- ❖ CNS disturbances occasionally

Adverse Effects

- Trimethoprim **inhibits secretion of creatinine**
 - mild elevation of serum creatinine without impairment of GFR.
 - important to distinguish from true nephrotoxicity that may be caused by sulfonamides.

QUINOLONES

Introduction

➤ Nalidixic acid: the first quinolone

- Used for many years in treatment of UTIs

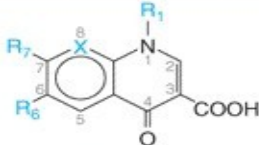
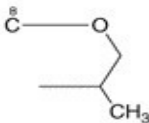
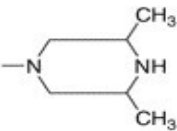
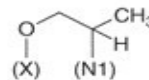
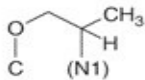
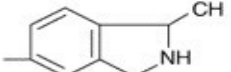
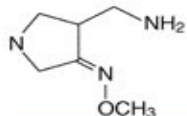
➤ Fluorinated 4-quinolones

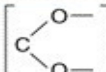
- clinically relevant quinolones are synthetic **fluorinated analogs of nalidixic acid**
- Broad antimicrobial activity
- Effective after oral administration for the treatment of a wide variety of infectious diseases
- Relatively few side effects
- Microbial resistance does not develop rapidly

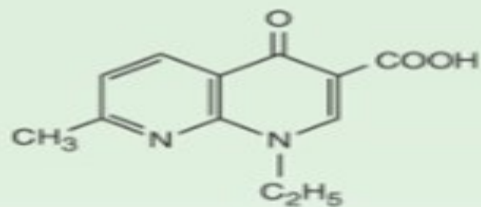
Fluoroquinolones

- ❖ They are active against a variety of Gram-positive and Gram-negative bacteria.
- ❖ They are also active against agents of
 - atypical pneumonia**
 - Intracellular pathogens such as Legionella and
 - some mycobacteria, including MTB and Mycobacterium avium complex.

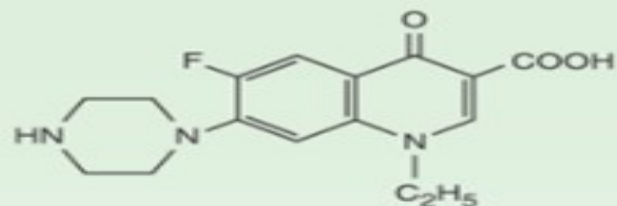
Table 43-2. Structural Formulas of Selected Quinolones and Fluoroquinolones

				
CONGENER	R ₁	R ₆	R ₇	X
Nalidixic acid	—C ₂ H ₅	—H	—CH ₃	—N—
Cinoxacin (<i>N</i> replaces C2)	—C ₂ H ₅	(Fused dioxolo ring)*		—CH—
Norfloxacin	—C ₂ H ₅	—F		—CH—
Ciprofloxacin		—F		—CH—
Ofloxacin		—F		—CH—
Sparfloxacin (—NH ₂ on C5)		—F		
Fleroxacin	—CH ₂ —CH ₂ —F	—F		
Pefloxacin	—C ₂ H ₅	—F		—CH—
Levofloxacin		—F		
Garenoxacin		—H		
Gemifloxacin		—F		—N—

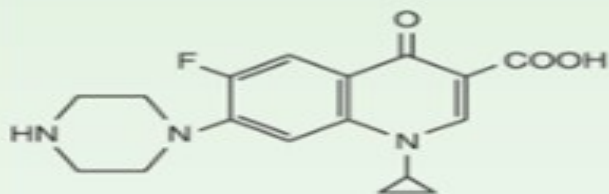
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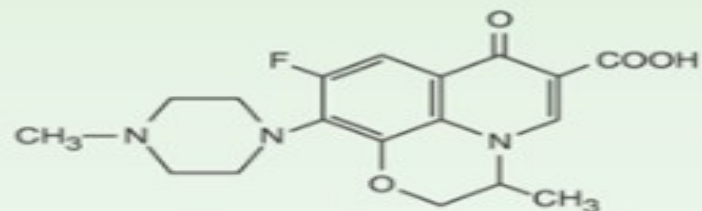
Nalidixic acid



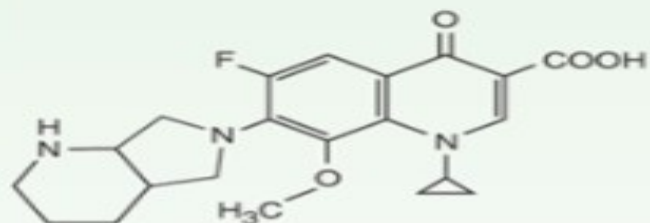
Norfloxacin



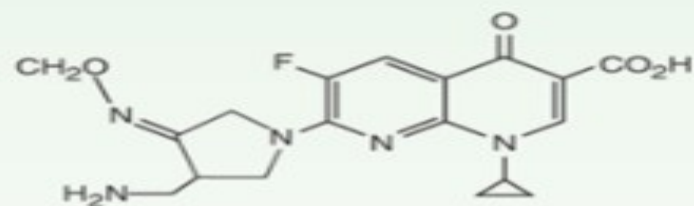
Ciprofloxacin



Levofloxacin



Moxifloxacin



Gemifloxacin

Mechanism of Action

- ❖ Quinolones block bacterial DNA synthesis by inhibiting **bacterial topoisomerase II** (DNA gyrase) and **topoisomerase IV**.
- ❖ DNA gyrase.....helps relaxation of positively supercoiled DNA that is required for normal **transcription** and replication.
- ❖ Topoisomerase IV.....for **separation of replicated** chromosomal DNA into the **respective daughter cells during cell division**.

Table: Pharmacokinetic properties of some fluoroquinolones.

Drug	Half-Life (h)	Oral Bioavailability (%)	Peak Serum Concentration (mcg/mL)	Oral Dose (mg)	Primary Route of Excretion
Ciprofloxacin	3-5	70	2.4	500 twice daily	Renal
Gemifloxacin	8	70	1.6	320 once daily	Renal and nonrenal
Levofloxacin	5-7	95	5.7	500 once daily	Renal
Moxifloxacin	9-10	>85	3.1	400 once daily	Nonrenal
Norfloxacin	3.5-5	80	1.5	400 twice daily	Renal
Ofloxacin	5-7	95	2.9	400 twice daily	Renal

❖ **Norfloxacin**.....is the least active of the fluoroquinolones against both Gram (+) and(-), MICs 4to 8 fold higher than others.

Ciprofloxacin	}	excellent G(-) and moderate to good G(+)
Enoxacin,		
Lomefloxacin		
levofloxacin,		
ofloxacin, and		
pefloxacin		

❖ MSSA are susceptible but not MRSA

✓ In staphylococcal infections.....in combination with rifampin or other.....to **prevent resistance**

- ❖ **Enterococci** less susceptible than staphylococci,
- ❖ **Ciprofloxacin** is the most active agent against Gram-(-)
particularly **P aeruginosa**.
- ❖ **Levofloxacin**, has superior activity against Gram (+)
 - Especially **Streptococcus pneumoniae**.

Gatifloxacin, gemifloxacin, and moxifloxacin

- ❖ Have improved activity against Gram (+), particularly **S pneumoniae** and **some staphylococci**
- ❖ G (-) Activity < **Ciprofloxacin**
- ❖ Have less MICs for staphylococci compared to ciprofloxacin group
- ❖ Moxifloxacin has modest activity against **anaerobic** but not against **P aeruginosa**.
- ❖ Moxifloxacin is Mostly used as **ophthalmic** preparation

Resistance

resistance is due to

- ✓ one or more point mutations in the quinolone binding region
- ✓ to a change in the permeability of the organism..
- ✓ Resistance to one fluoroquinolone...cross-resistance to all other members of this class.

Pharmacokinetics

- ❖ The relatively long half-lives of levofloxacin, gemifloxacin, and moxifloxacin
 - Dose..... once-daily dosing.
- ❖ absorption is impaired by divalent and trivalent cations,
- ❖ be given 2 hours **before** or 4 **hours** after these products
- ❖ Most fluoroquinolones eliminated by renal mechanisms
- ❖ Dosage adjustment for $\text{CrCL} < 50 \text{ mL/min}$,
 - Except **moxifloxacin** and **pefloxacin** since it is metabolized in the liver
 - Be given with caution in patients with hepatic failure

Clinical Uses

- ❖ Except moxifloxacin used for UTI
 - Efficacy in UTI (fluro >>TMP-SMX)
- ❖ Diarrhea
- ❖ infections of soft tissues, bones, and joints
- ❖ intra-abdominal and respiratory tract infections
- ❖ **Ciprofloxacin a choice** for prophylaxis and treatment of anthrax;
- ❖ levofloxacin is also approved by FDA for prophylaxis.

Clinical Uses...

- ❖ Azithromycin + gemifloxacin
 - for the treatment of **gonococcal infection** an alternate to ceftriaxone
- ❖ Levofloxacin and ofloxacin
 - as alternative options for **chlamydial urethritis** or **cervicitis**.
- ❖ Ciprofloxacin, levofloxacin, or moxifloxacin
 - part of a treatment regimen for TB and Non-TB
 - for eradication of meningococci from carriers and for
 - prophylaxis in neutropenic cancer patients.

Clinical Uses...

Respiratory fluoroquinolones

- for treatment of lower respiratory tract infections.

Adverse effects

- ❖ generally well tolerated.
- ❖ The most common effects are..... nausea, vomiting, and diarrhea.
- ❖ gatifloxacin, levofloxacin, gemifloxacin, and moxifloxacin
 - Prolongation of the QTc interval
- ❖ Gatifloxacin hyperglycemia in DM patients hypoglycemic agents.
- ❖ Quinolones can **damage growing cartilage** and **Tendinitis**
- ❖ Neuropathy

ANTITUBERCULOSIS DRUGS

- Characteristics of mycobacteria that make the diseases chronic & necessitate prolonged treatment
 - Mycobacteria grow slowly and may be dormant in the host for long periods
 - Many antibacterials do not penetrate mycobacterial cell walls, and a portion reside inside macrophages
 - Mycobacteria are agile in developing resistance to single chemotherapeutic agents
- Consequence ⇒ effective therapy requires a prolonged course of multiple drugs
 - Patient compliance, drug toxicity, drug interactions, concurrence of other diseases

Tuberculosis

- Two species of *Mycobacterium* cause tuberculosis:
M. tuberculosis and *M. bovis*
- *M. tuberculosis* is transmitted by inhalation of infective droplets coughed or sneezed into the air by a patient with tuberculosis.
- *M. bovis* is transmitted by milk from diseased cows and first produces intestinal or tonsillar lesions.

Drugs for Tuberculosis

➤ *First-line drugs are superior in efficacy and possess an acceptable degree of toxicity.*

- Include isoniazid, rifampin, pyrazinamide, ethambutol, and streptomycin.
- Most patients with tuberculosis can be treated successfully with these drugs.

➤ *Second-line drugs are more toxic and less effective*

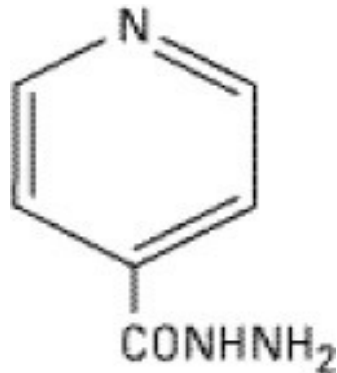
- Indicated only when the *M. tuberculosis* organisms are resistant to the first-line agents.
- Include cycloserine, ethionamide, aminosalicylic acid, rifabutin, quinolones, capreomycin, viomycin, and thiacetazone.

First-line drugs for tuberculosis

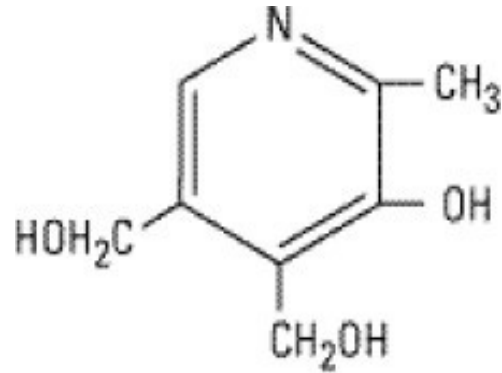
Isoniazid (isonicotinic acid hydrazide, INH)

➤ Chemistry

- Isoniazid is the hydrazide of isonicotinic acid
- Structurally similar to pyridoxine
- A small, simple molecule freely soluble in water



Isoniazid



Pyridoxine

Antibacterial Activity

- The most active drug for the treatment of tuberculosis caused by susceptible strains
- Is **bacteriostatic** for "resting" bacilli, but is **bactericidal** for rapidly dividing microorganisms
- Remarkably selective for mycobacteria
- Penetrates cells with ease and is just as effective against bacilli growing within cells as it is against those growing in culture media (**Effective intracellular activity**)

Mechanism of Action

- Mycobacterial **catalase-peroxidase** (*katG* encoded) converts isoniazid into an active metabolite
- A primary action of isoniazid is to **inhibit the biosynthesis of mycolic acids (MA)**
- Catalase-peroxidase-activated isoniazid binds enoyl-ACP reductase of fatty acid synthase II,
 - which converts Δ^2 -unsaturated fatty acids to saturated fatty acids in the mycolic acid biosynthetic pathway.
- It also inhibits mycobacterial catalase-peroxidase, which may **increase the likelihood of damage to the mycobacteria from reactive oxygen species and H_2O_2**

Pharmacokinetics

- Water soluble, well absorbed from oral & parenteral sites
- Does not bind to serum proteins, diffuses readily into all body fluids and cells
- Concentrations in CNS & CSF are similar to those in plasma during meningeal inflammation
- INH is acetylated to acetyl INH by *N-acetyltransferase*
 - Genetic polymorphism in the rate of acetylation.
 - ✓ Slow or rapid acetylation is rarely important clinically, although slow inactivators tend to develop peripheral neuropathy more readily.

Clinical Uses

- Isoniazid is the safest and most active mycobactericidal
- Considered the primary drug for use in all therapeutic and prophylactic regimens for susceptible tuberculosis infections
- As a single agent for treatment of latent tuberculosis, which is usually determined by a positive tuberculin skin test.

Adverse Reactions

➤ Allergic Reactions

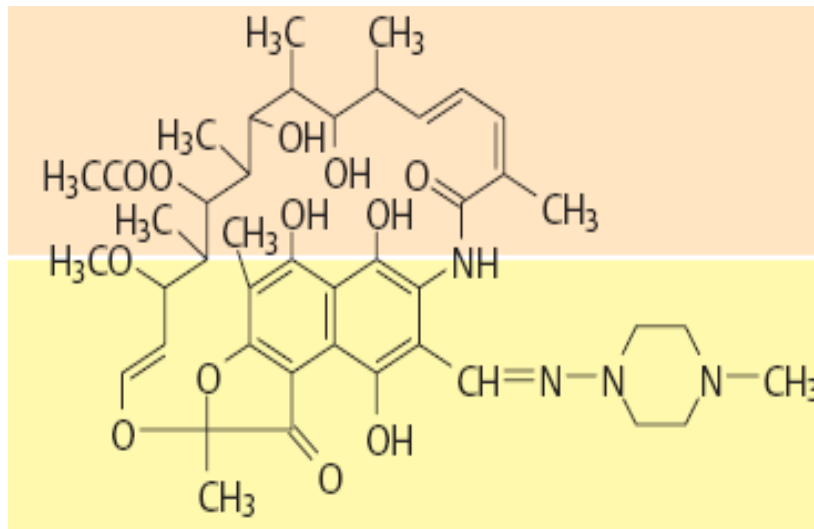
- Fever, skin rashes

➤ Direct Toxicity

- **Hepatitis** (greater risk in alcoholics & pregnancy and is a contraindication to further use of the drug).
- **Peripheral neuropathy** due to a relative pyridoxine deficiency (more likely in slow acetylators, malnutrition, alcoholism, diabetes, AIDS)
 - ✓ Minimized by administration of **pyridoxine**
- **CNS toxicity** (memory loss, psychosis, and seizures)
 - ✓ May respond to **pyridoxine**

Rifampin and other rifamycins

- The rifamycins (rifampin, rifabutin, rifapentine) are a group of structurally similar, complex macrocyclic antibiotics produced by *Ammycolatopsis mediterranei*
- Rifampin is a semisynthetic derivative of rifamycin B.
- Rifampin is soluble in organic solvents and in water at acidic pH.



Antibacterial Activity

- Rifampin inhibits the growth of *M. tuberculosis* *in vitro*.
- *It also inhibits the growth of nontuberculous mycobacteria*
- Rifampin inhibits the growth of most
 - Gm(+) bacteria
 - Gm(-) microorganisms such as *E. coli*, *Pseudomonas*, *Proteus*, and *Klebsiella*.

Mechanism of Action

- Rifampin binds DNA-dependent RNA polymerase → forms stable drug-enzyme complex → leading to suppression of initiation of chain formation in RNA synthesis.
- Nuclear eukaryotic RNA polymerases are unaffected
- High concentrations can inhibit
 - RNA synthesis in mammalian mitochondria,
 - viral DNA-dependent RNA polymerases, and
 - reverse transcriptases.
- **Bactericidal for both intracellular and extracellular**

Bacterial Resistance

➤ Due to an alteration of the DNA-dependent RNA Polymerase

- Reduced binding of the drug to the polymerase
- ✓ It should not be used alone since resistance develops very fast

Pharmacokinetics

- Well absorbed after **oral** administration and excreted mainly through the liver into bile.
- **Undergoes enterohepatic recirculation**
- Intestinal reabsorption is **reduced** by **food**
- Small amount excreted in urine
- **Distributed widely** in body fluids and tissues
- Rifampin is relatively **highly protein-bound**,

Clinical Uses

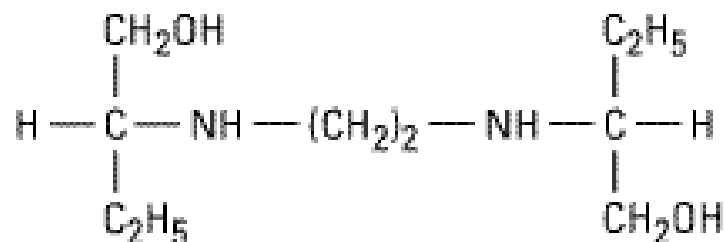
- A first-line antitubercular drug used in the treatment of all forms of **pulmonary** and **extrapulmonary TB**
- Alternative to INH in the treatment of latent TB infection
- In combination with antileprosy agent for the treatment of leprosy
- *Also used in **MRSA** infections*

Adverse Reactions

- Imparts orange color to urine, sweat, tears
- Rashes, thrombocytopenia, and nephritis, cholestatic jaundice and occasionally hepatitis
- It is a strong inducer of most cytochrome P450 isoforms
 - Increases the elimination of numerous drugs including methadone, anticoagulants, some anticonvulsants, protease inhibitors, and

Ethambutol

- Ethambutol is a synthetic, water-soluble, heat-stable compound



Ethambutol

Antibacterial Activity

- Nearly all strains of *M. tuberculosis* and *M. kansasii* as well as a number of strains of MAC are sensitive
- Ethambutol has no effect on other bacteria.
- **It suppresses the growth of most INH-and streptomycin-resistant tubercle bacilli.**
- Resistance to ethambutol develops very slowly *in vitro*.

Mechanism of Action and Resistance

- Inhibits **arabinosyl transferases** involved in cell wall biosynthesis.
- The enzyme is involved in **the polymerization reaction of arabinoglycan**, an essential component of the mycobacterial cell wall

Pharmacokinetics

- Well absorbed from the GIT
- About **20% of the drug is excreted in feces and 50% in urine in unchanged form** (dose adjustment in renal impairment)
- Is widely distributed in all body fluids

Clinical Uses

- Has replaced aminosalicylic acid as a first-line antitubercular drug.
- Commonly included as a 4th drug, along with INH, pyrazinamide, and rifampin, in patients infected with MDR strains.
- Also used in combination in the treatment of *M. aviumintracellulare* infection in AIDS patients.

Adverse Reactions

- Hypersensitivity, loss of visual acuity and red-green color blindness (most serious side-effects)
- Ethambutol is relatively contraindicated in children too young to permit assessment of visual acuity and red-green color discrimination.

Streptomycin

- An aminoglycoside antibiotic and was the first drug to reduce tuberculosis mortality.
- It is **bactericidal** against *M. Tuberculosis* in vitro but is **inactive** against **intracellular** organisms.
- Most *M. tuberculosis* strains and nontuberculosis species, such as *M. kansasii* and *M. Aviumintracellulare*, are sensitive.

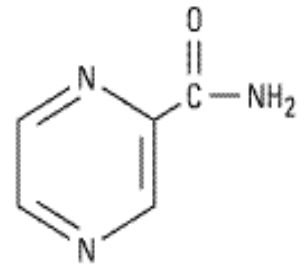
Antibacterial Activity

- It penetrates into cells poorly, and is active mainly against **extracellular** tubercle bacilli

Clinical Use in TB

- **As a fourth drug in combination with INH, rifampin, and pyrazinamide in patients at high risk for drug resistance**
- **Employed when an injectable drug is needed or desirable, (severe, possibly life-threatening forms of TB, and in treatment of infections resistant to other drugs)**

Pyrazinamide



Pyrazinamide (PZA)

- Synthetic pyrazine analog of nicotinamide
- **At neutral pH, it is inactive in vitro, but at pH 5.5 it inhibits tubercle bacilli**
- **Drug is taken up by macrophages and exerts its activity against intracellular organisms residing within acidic environment.**

Mechanism of Action

- Exact MOA unknown, although its target appears to be the **mycobacterial fatty acid synthetase** involved in **mycolic acid biosynthesis**.
- It is a **prodrug** activated to the active **pyrazinoic acid** by the mycobacterial enzyme **pyrazinamidase**

Pharmacokinetics

➤ Pharmacokinetics

- Well absorbed from the GIT and widely distributed in body tissues, including inflamed meninges

➤ Clinical uses

- An important front-line drug used in conjunction with INH and rifampin in short-course (ie, 6-month) regimens as a "**sterilizing**" agent active against residual intracellular organisms that may cause relapse.

Adverse effects

- **Hepatotoxicity** (in 1-5%), nausea, vomiting, drug fever, and hyperuricemia

Antiretroviral Agents

- HIV is a chronic viral infection with no cure
- HIV primarily affects the CD4 lymphocyte
- When CD4 cells are destroyed, a person's immunity is impaired
- As immune function decreases, **opportunistic** infections increase
- HIV progresses over time to death, if no treatment is given

- HIV infects cells that express **CD4 receptor molecules**
 - T4-lymphocytes (T-helper cells)
 - Monocyte-macrophage cell lines
- T4-lymphocytes are type of white blood cell that ‘switch on’ the immune system to fight disease
- HIV uses CD4 cells for reproduction

- Successful entry of the virus to a target cell also requires **cellular co-receptors**
- A fusion co-receptor is designated CXCR4 for **T-cell tropic strain** and CCR5 for **monocyte-macrophage tropic strains**
- The receptor and co-receptors of CD4 cells interact with HIV's gp-120 and gp-41 proteins during entry into a cell

Drugs

- 1) Nucleoside Reverse Transcriptase Inhibitors (NRTIs)
- 2) Nonnucleoside Reverse Transcriptase Inhibitors (NNRTIs)
- 3) Protease inhibitors
- 4) Entry inhibitors
- 5) Integrase inhibitors
- 6) CCR5 receptor antagonist: Maraviroc

Classification

1. Reverse Transcriptase Inhibitors

A. Nucleoside/Nucleotide Reverse Transcriptase Inhibitors

(NRTIs/NtRTIs)

- ✓ Zidovudine (ZDV)*
- ✓ Didanosine (ddi)*
- ✓ Lamivudine (3TC)*
- ✓ Zalcitabine (ddC)
- ✓ Stavudine (d4T)*
- ✓ Abacavir (ABC)*
- ✓ Emtricitabine (FTC)

Classification (cont.)

- B. Non-Nucleoside Reverse Transcriptase Inhibitors (NNRTIs)
 - Nevirapine (NVP)*
 - Delavirdine (DLV)
 - Efavirenz (EFV)*

Classification Cont.

2. Protease Inhibitors (PIs)

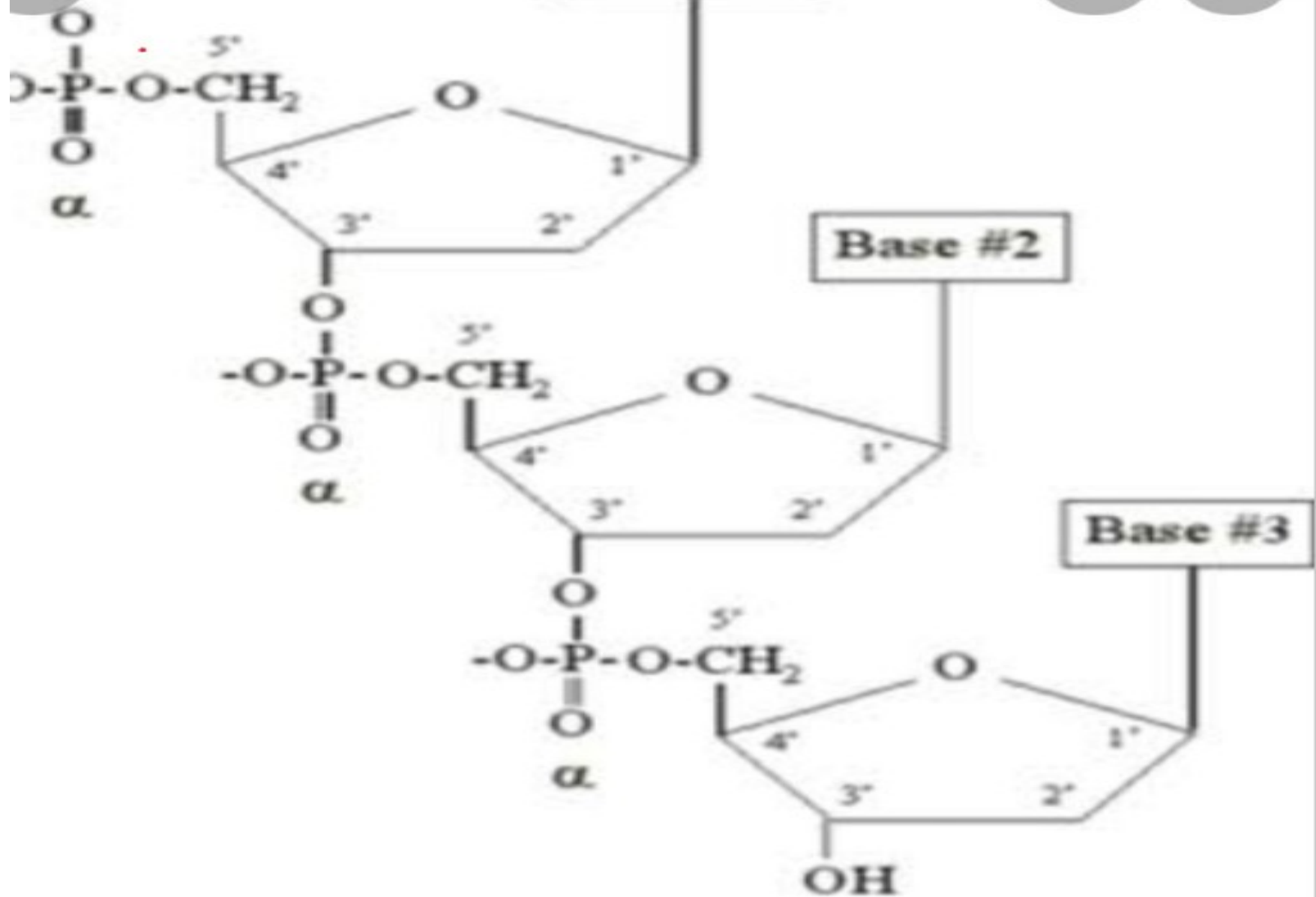
- ✓ Saquinavir (SQV)*
- ✓ Ritonavir (RTV)*
- ✓ Indinavir (IDV)*
- ✓ Nelfinavir (NFV)*
- ✓ Amprenavir (APV)
- ✓ Fos-amprenavir (f-APV)
- ✓ Lopenavir/ritonavir (LPV/r)*
- ✓ Atazanavir (ATZ)

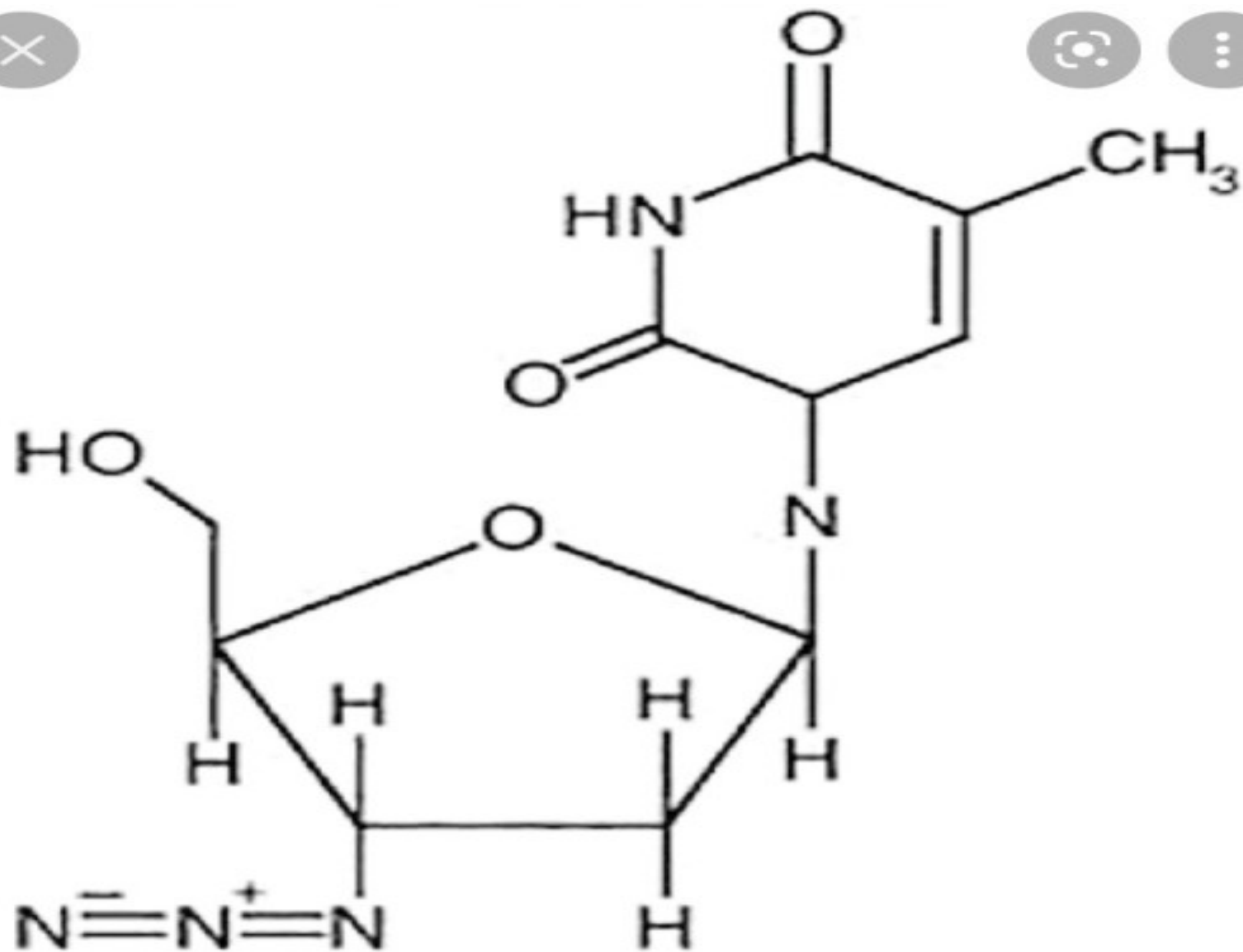
3. Fusion inhibitors

- ✓ Enfuvirtide (T-20)

Chemistry of NRTIS

NRTI	Nucleoside/ Nucleotide base
Zidovudine	thymine
Didanosine	adenine
Lamivudine	
Zalcitabine	cytosine
Stavudine	
Abacavir	cytosine
Emtricitabine	cytosine
	thymine
*Tenofovir disoproxil	adenine
fumarate	guanine
* Adenine mono phosphate	

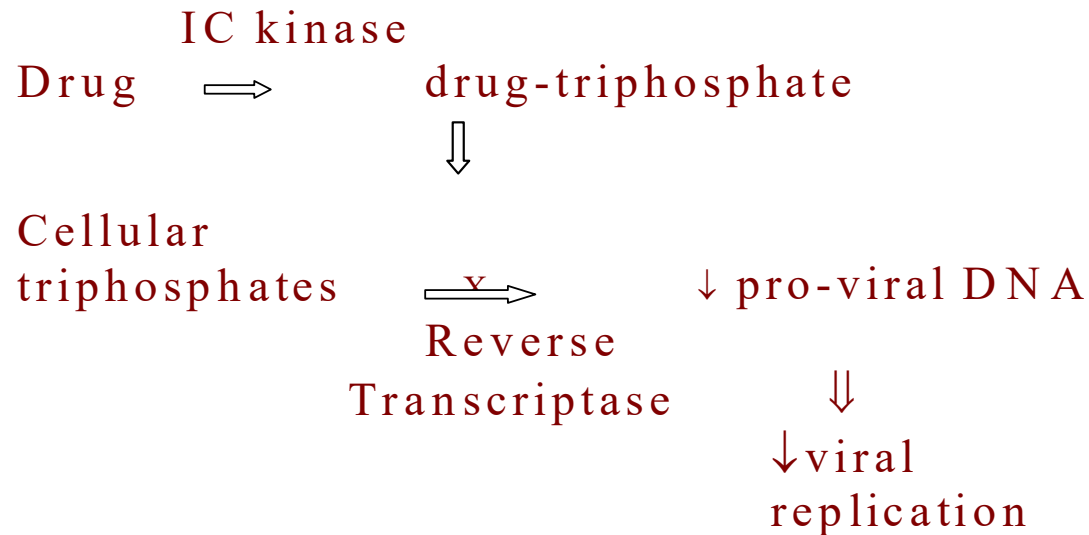




NRTIs Used to Treat HIV Infection

- NRTIs are analogs of native ribosides (nucleosides or nucleotides containing ribose), which lack a 3'-OH group
- **Mechanism of action:** They are phosphorylated by cellular enzymes to corresponding TPes which are incorporated into **viral DNA by virus reverse transcriptase.**
- Because the **3'-hydroxyl group is not present**, a 3'-5'-phosphodiester bond between an incoming nucleoside TPes and the growing DNA chain cannot be formed, and **DNA chain elongation is terminated**
- The drugs have lower affinities for **host cell DNA polymerases**
- although **mitochondrial DNA polymerase appears to be susceptible** at therapeutic concentrations)

Mechanism of action



N.B. HIV reverse transcriptase is 20-30 times more susceptible than alpha DNA polymerase of the mammalian cells, but gamma DNA polymerase is equally susceptible

- **Pharmacokinetics:** primarily renally excreted, & require dosage adjustment in renal insufficiency
- **Adverse effects:** Many of the toxicities of the NRTIs are believed to be due to **inhibition of the mitochondrial DNA polymerase** in certain tissues.
- **Dideoxynucleosides** (zalcitabine, didanosine, & stavudine) have a greater affinity for **mitochondrial DNA polymerase**
 - leading to **peripheral neuropathy, pancreatitis, and lipoatrophy**
- **Avoid overlapping toxicities when > one NRTI is given**
- **Drug interactions:** Not many
- **Cross-resistance** occurs between agents of the same analog class (thymidine, cytosine, guanosine and adenosine)
 - so **concomitant use of agents** in the same class is contraindicated (for example, **zidovudine plus stavudine**).

A. Zidovudine (AZT)

- the first agent available for treatment of HIV infection
- approved for use **in children and adults and to prevent prenatal infection in pregnancy**
- recommended **for prophylaxis in those exposed to HIV**
- The drug is **well absorbed after oral administration.**
- Penetration **across the BBB is excellent, and the drug has a half-life of 1 hour.**
- AZT is toxic **to bone marrow.** Headaches are common.
- Both **stavudine** and **ribavirin** are activated by the same intracellular pathways and should not be given with AZT.

B. Stavudine (d4T)

- d4T is an analog of **thymidine**, in which a double bond joins the 2' and 3' carbons of the sugar.
- strong **inhibitor of cellular enzymes (DNA polymerases)**, ↓ing **mitochondrial DNA synthesis**, resulting in toxicity
- The major and most common clinical toxicity is **peripheral neuropathy along with lipoatrophy and hyperlipidemia**.

C. Tenofovir (TDF)

- It is converted by **cellular enzymes to the diphosphate**, which is the **inhibitor of HIV reverse transcriptase**.
- some AZT- **resistant** strains retain susceptibility to **tenofovir**.
- **Taken with a meal to increase bioavailability.**
- Tenofovir has a **long half-life**, allowing **once-daily dosing**.
- Most of the drug is **recovered unchanged in the urine**
- **Serum creatinine must be monitored** and doses adjusted in renal insufficiency.
- **GI complaints** are frequent and include NDV
- **The only NRTI with significant drug interactions.**
- **Tenofovir** decreases the concentrations of **atazanavir** such that atazanavir must be boosted with ritonavir

D. Lamivudine (3TC)

- 3TC is approved for **treatment of HIV in combination with AZT**
- 3TC terminates the synthesis of the **proviral DNA chain**, and **inhibits the reverse transcriptase of both HIV & HBV**.
- However, **it does not affect mitochondrial DNA synthesis or bone marrow precursor cells**.
- It has **good bioavailability on oral administration**, depends on the kidney for excretion, and is well tolerated.

E. Emtricitabine (FTC)

- FTC, a fluoro-derivative of **lamivudine**, **inhibits both HIV and HBV reverse transcriptase**.
- **FTC is orally active**, with a mean **bioavailability of 93 %**
- It is eliminated essentially **unchanged** in the urine
- **Headache**, diarrhea, nausea, rash: common adverse effects

F. Abacavir (ABC)

- ABC a **guanosine** analog. There may be some **cross-resistance** with strains resistant to **AZT and lamivudine**.
- It is **well absorbed orally, & metabolites appear in the urine**
- Common side effects include **gastrointestinal disturbances, headache, and dizziness**.
- Approximately **5% of patients exhibit hypersensitivity reaction which is characterized by drug fever, plus one or more of the following symptoms of:**
 - **Rash**
 - gastrointestinal symptoms
 - malaise, and
 - respiratory distress

Nonnucleoside Reverse Transcriptase Inhibitors (NNRTIs)

- **Nevirapine (NVP)**
- **Delavirdine (DLV)**
- **Efavirenz (EFV)**

Mechanism of Action *NNRTIs*

- Bind to site on viral reverse transcriptase, different from NRTIs
 - results in blockade of RNA and DNA dependent DNA polymerase activity
 - do **not** compete with nucleoside triphosphates
 - do **not** require phosphorylation
- these drugs can **not** be given alone
- substrates and inhibitors of CYP3A4

NNRTIs Used to Treat AIDS

- NNRTIs are **highly selective, noncompetitive inhibitors of HIV-1 reverse transcriptase.**
- Bind to HIV reverse transcriptase adjacent to the active site, induce **conformational change and enzyme inhibition.**
- Major advantage: their lack of effect on the **host blood-forming elements and cross-resistance with NRTIs.**
- These drugs, however, do have common characteristics that include:
 - cross-resistance within the NNRTI class
 - drug interactions, and
 - a high incidence of hypersensitivity reactions, including rash.

A. Nevirapine (NVP)

- Used in combination with other antiretroviral drugs **for the treatment of HIV-1 infections in adults and children.**
- Severe **hepatotoxicity**
 - NVP should not be initiated with **CD4+ T-cell counts greater than: in ♀ 250 cells/mm³ or in ♂ 400 cells/mm³.**
- NVP is well absorbed orally
- **NVP is an inducer of the CYP3A4 increasing metabolism of oral** contraceptives, ketoconazole, methadone, metronidazole, quinidine, theophylline, and warfarin.
- SE's: **rash, fever, headache, and fatal hepatotoxicity,**

B. Efavirenz (EFV)

- **EFV treatment results in ↑in CD4+ and a ↓in viral load comparable to that achieved by protease inhibitors when used in combination with NRTIs; (it is preferred NNRTI)**
- **Bioavailability is ↑when taken with a high-fat meal.**
- **A half-life of more than 40 hours**
- **Most adverse effects are associated with CNS: dizziness, headache, vivid dreams, and loss of concentration**
- **Rash is the other most common side effect, with an incidence of approximately 25 percent.**

Nonnucleoside Reverse Transcriptase Inhibitors (NNRTIs)



- **Nevirapine-** prevents transmission of HIV from mother to newborn when given at onset of labor and to the neonate at delivery

Protease Inhibitors

- **Indinavir (IDV)**
- **Ritonavir (RTV)**
- **Saquinavir (SQV)**
- **Nelfinavir (NFV)**
- **Lopenavir (LPVr)**
- **Atazanavir (ATV)**

Protease Inhibitors

- The **protease enzyme cleaves precursor molecules to produce mature, infectious virions**
- **inhibit protease and prevent the spread of infection**
- **cause a syndrome of altered body fat distribution, insulin resistance, and hyperlipidemia.**
- Inhibitors of HIV protease have significantly altered the course of this devastating viral disease

- **Mechanism of action:** reversibly inhibit the HIV aspartyl protease, responsible for cleavage of the viral polyprotein into a number of essential enzymes (reverse transcriptase, protease, and integrase) and several structural proteins.
- Exhibit at least a **thousand-fold greater affinity for HIV-1 and HIV-2 enzymes** than for comparable human proteases, such as renin and cathepsin (selective toxicity)

- **Treatment with a PI and two NRTIs results in a decrease in the plasma viral load to undetectable levels** in 60 to 95 percent of patients.
- **Pharmacokinetics**: have **poor oral bioavailability**.
- Extensive metabolism, little excreted unchanged in urine
- **Adverse effects**: PIs commonly cause parathesias, NVD
- Disturbances in glucose & lipid metabolism (diabetes, **hypertriglyceridemia, and hypercholesterolemia**) occur

- **Drug interactions:** a common problem for all PI's; they are substrates and **potent inhibitors of CYP isozymes.**
- Drugs that rely on metabolism & contraindicated:-
 - **rhabdomyolysis from simvastatin or lovastatin**
 - **excessive sedation from midazolam** and
 - respiratory depression from fentanyl
- **D/Is** that require dosage modification and cautious use with PI's :warfarin, sildenafil and phenytoin
- **CYP Inducers**→(may) lower [PI]→treatment failures
- Thus, drugs such as **rifampin** and St. John's wort are also contraindicated with protease inhibitors.

A. Ritonavir (RTV)

- **RTV is no longer used as a single PI** but, instead, is used as a PK enhancer or booster of other PI's.
- **Ritonavir** is a potent inhibitor of CYP3A, and concomitant ritonavir administration (at low doses) increases the **bioavailability of the second PI** → longer dosing intervals
- **Ritonavir** is also a **self-inducer of its own metabolism**

B. Saquinavir (SQV)

- To maximize bioavailability, SQV can be given along with a low dose of RTV.
- **High-fat meals** also enhance absorption.
- **Drugs that enhance the metabolism of SQV**, such as rifampin, NVP, EFV... avoided if possible
- AEs: headache, fatigue, ND & GI disturbances.

C. Indinavir (IDV)

- IDV is well **absorbed orally** and, of all the PI's, is **the least protein-bound**, at **60 %**.
- **Acidic gastric conditions** are necessary for absorption
- Indinavir has the **shortest half-life of the PI's**, at **1.8 hours**.
- It is **well tolerated**, with **usual GI symptoms** & headache)
- **Indinavir** characteristically causes **nephrolithiasis** (adequate hydration is important) and **hyperbilirubinemia**.
- **Fat redistribution** is particularly worrying with this drug.

Fusion Inhibitors

- *Enfuvirtide* (T-20)- binds to the **gp41 subunit** of the **viral envelope glycoprotein**, preventing the **conformational changes required for fusion** of the viral and cellular membranes
- By blocking fusion (entry into cell), prevents HIV from infecting CD4 cells

Integrase inhibitor-Raltegravir

- Inhibits the viral enzyme integrase
- HIV Integrase nicks the host chromosomal DNA and integrates the proviral DNA with it
- Active against both HIV 1 and 2 and causes improved CD4 cell count
- Uses:
 - As a component of initial triple drug regimen along with 2NRTIs
- Adverse effect: myopathy

Combining NRTIs

Strongly recommended

ABC + 3TC + EFV (or NVP)

ABC + 3TC + LPV/r

AZT + 3TC + LPV/r

TDF+3TC+DTG

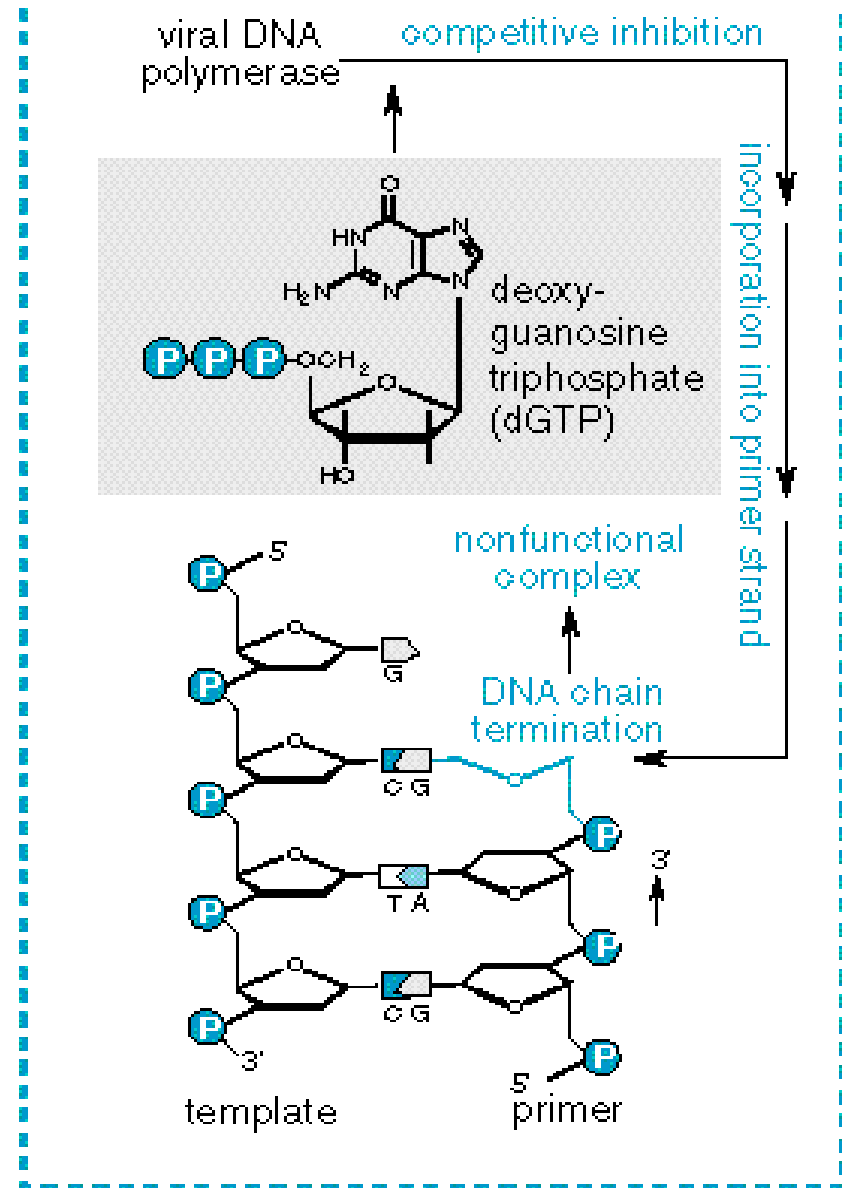
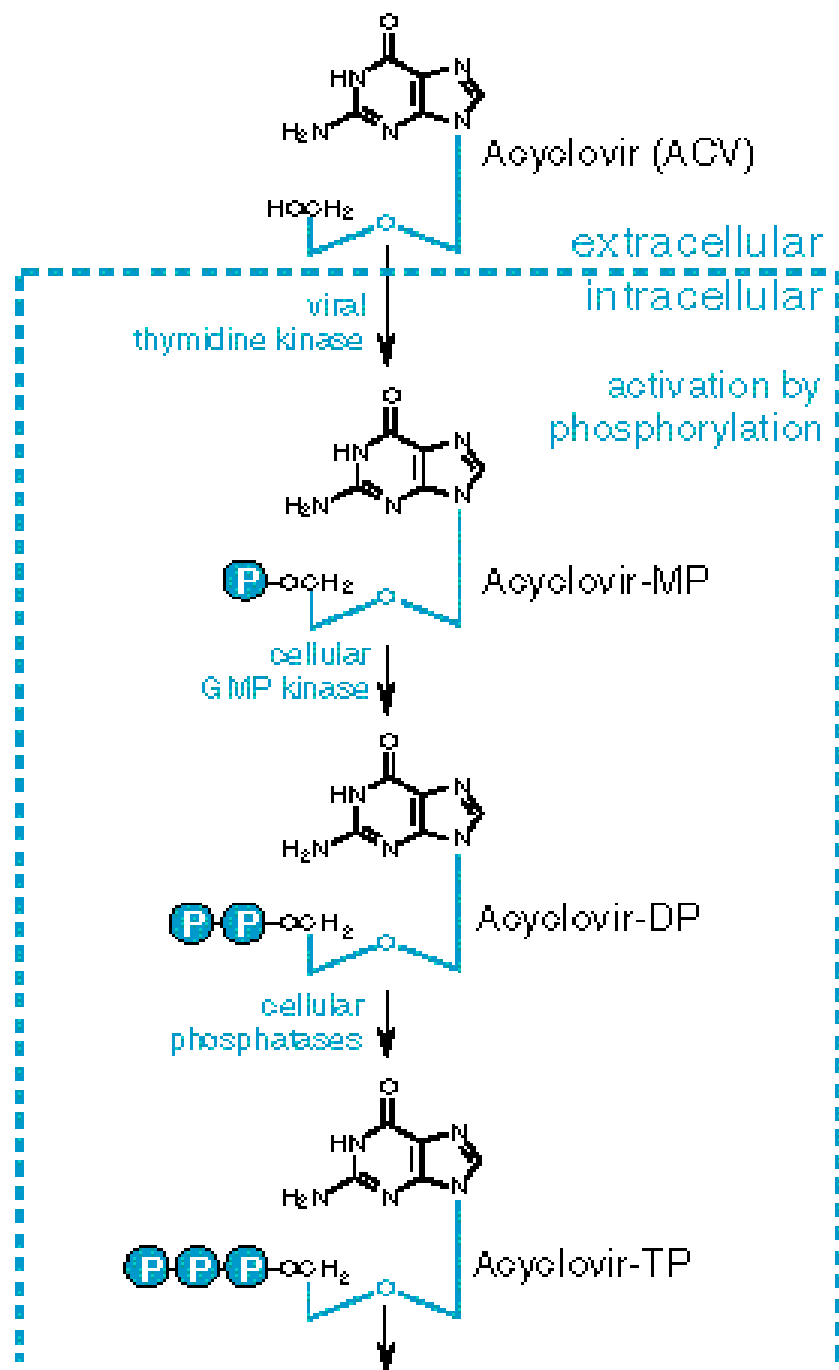
1. Antiherpes Agents

- *Acyclovir- prototype*
- Valacyclovir
- Famciclovir
- Penciclovir
- Docosanol
- Trifluridine
- Vidarabine

Mechanism of Action

Acyclovir

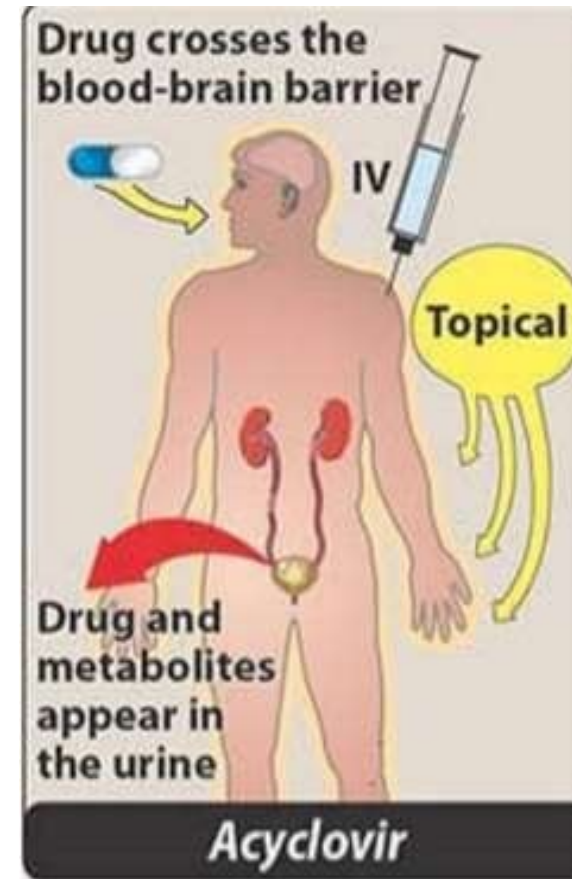
- an acyclic guanosine derivative
- Phosphorylated by *viral thymidine kinase*
- Di-and tri-phosphorylated by host cellular enzymes
- Inhibits viral DNA synthesis by:
 - 1) competing with dGTP for viral DNA polymerase
 - 2) chain termination



Acyclovir

❖ Pharmacokinetics:

- About 20% of an oral dose of acyclovir is absorbed
- Less plasma protein bound
- Primarily excreted unchanged in **urine** both by **glomerular** filtration and tubular secretion
- It accumulates in patients with renal failure
- Its plasma half life is 2-3 hours



Clinical Uses

Acyclovir

- **Uses:**
 - Herpes Simplex Virus 1 and 2 (HSV)
 - Oral acyclovir is **effective for treatment of primary infection and recurrences of genital and labial herpes.**
 - Intravenous acyclovir is the treatment of choice for herpes simplex **encephalitis, neonatal HSV infection** and
 - for severe primary, recurrent **HSV genital and labial infections** and for those who cannot ingest oral pills
 - Varicella-zoster virus (VZV)

Clinical Uses

Acyclovir

- ***Side Effects***

- Depends on the route of adm.

- IV= renal dysfunction, neurotoxicity (tremors, and delirium)
 - Oral = diarrhea and headache
 - Topical = local irritation

Valacyclovir

- L-valyl ester of acyclovir
- **Converted to acyclovir when ingested**
- Improved oral bioavailability due to active transport by peptide transporters in the intestine
- 3-5 x BA than Acyclovir
- **M.O.A.:** same as acyclovir
- **Uses:**
 - 1) recurrent genital herpes
 - 2) herpes zoster infections
- **Side Effects:** nausea, diarrhea, and headache

Famciclovir

- Prodrug of **penciclovir** (an acyclic guanosine analog)
- **M.O.A.:** same as acyclovir
 - does **not** cause chain termination
- **Uses:** Used as an alternative to acyclovir for genital or orolabial herpes and herpes zoster
- HSV-1, HSV-2, VZV, EBV, and hepatitis B
 - Effective in tt of first and recurrent **genital herpes**, and for **acute zoster**
- **Side Effects:** nausea, diarrhea, and headache

Penciclovir

- Guanosine analog, active metabolite of famciclovir
- Available for topical use
 - For recurrent *Herpes labialis* in immunocompetent adults
- Side effects are uncommon

Ganciclovir

- An analogue of acyclovir
- **requires triphosphorylation for activation**
- monophosphorylation is catalyzed by a **phosphotransferase** in CMV and by **thymidine kinase** in HSV cells.
- **Uses:** CMV* (100 x Acyclovir), HSV, VZV, and EBV
- **Note:** Used only for **prophylaxis and treatment of severe CMV infections** in immunocompromised patients
- ❖ **Adverse effects:**
 - Systemic toxicity is high (**bone marrow toxicity**, rash, fever, vomiting, neuropsychiatric disturbances)

Trifluridine

- Trifluridine-incorporates into viral and host DNA
 - Prevents it's systemic use
- **Uses:** HSV-1 and HSV-2 (topically)
t/t of primary **keratoconjunctivitis**
and recurrent keratitis.

Vidarabine

- An adenosine analog
- Inhibits **viral DNA polymerase**
- Incorporated into **viral and cellular DNA**
- Metabolized to hypoxanthine arabinoside
- **Side Effects: GI intolerance and myelosuppression**
- active against HSV-1, HSV-2, and VZV,
 - But, use limited to immunocompromised patients with herpetic and *vaccinial keratitis* and in HSV **keratoconjunctivitis**.

2. Anti-Cytomegalovirus Agents

- **Ganciclovir**
- **Valganciclovir**
- **Cidofovir**
- **Foscarnet**

Valganciclovir

- **L-valyl** ester prodrug of **ganciclovir**
- **M.O.A.:** same as gancyclovir
- **Uses:** CMV*
- **Side Effect:** myelosuppression

Cidofovir

❖ Mechanism of action:

- It inhibits viral DNA synthesis.
- Its phosphorylation is not dependent on viral enzymes (viral phosphokinase) and is converted to **the active diphosphate by cellular enzymes.**
- Cidofovir diphosphate does **not** preferentially **accumulate** in **virus infected cells.**

❖ Uses: Cidofovir

- CMV-induced **retinitis** in patients with AIDS

❖ Adverse effects:

- Dose related **kidney damage**
- Neutropenia, metabolic acidosis, uveitis and ocular hypotony also occur
- **Note:** Probenecid must be coadministered with cidofovir to reduce the risk of **nephrotoxicity**, but probenecid itself causes rash, headache, fever, and nausea

Foscarnet

- An inorganic pyrophosphate
- It **reversibly inhibits viral DNA and RNA polymerases**, and **HIV reverse transcriptase** without requiring phosphorylation.
- interfering with viral DNA and RNA synthesis
- **Uses:** HSV, VZV, CMV, EBV, HHV-6, HBV, and HIV
- Foscarnet is used for patients with **CMV retinitis** and **acyclovir-resistant HSV infection**
- Foscarnet has also been used to treat **CMV colitis** and esophagitis and acyclovir resistant VZV infection.

3. Anti-influenza virus

Amantadine

- Its antiviral activity is strain specific = inhibits replication of **influenza A** virus but not influenza B
- ❖ **Mechanism of action:**
 - It acts at **uncoating** as well as **viral assembly** in viral replication
 - Blocks the viral membrane matrix protein, M2, which functions as a channel for hydrogen ion
 - This channel is required for the fusion of the viral membrane with the cell membrane that ultimately forms the endosome

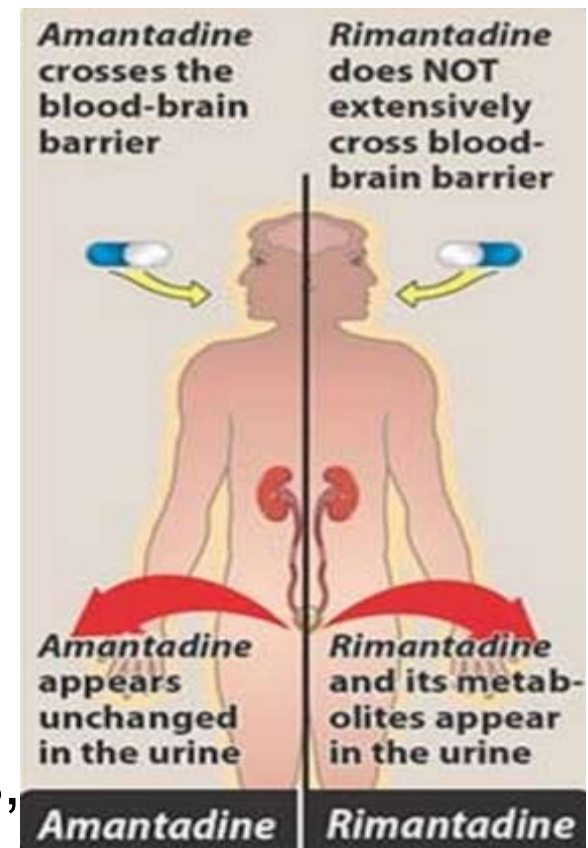
Amantadine

❖ Pharmacokinetics:

- Well absorbed orally and excreted unchanged in urine over 2-3 days

❖ Adverse effects:

- Nausea, anorexia, insomnia, dizziness, of mental concentration
- Hallucinations (rarely)
- Ankle edema (local vasoconstriction)



Amantadine

❖ **Uses:**

- **Prophylaxis of influenza A2** during an epidemic or seasonal influenza (~ 2 months)
- Treatment of influenzal (A2) illness
 - Reduction in fever, congestion, cough and quicker recovery
 - Parkinsonism

❖ **Contraindications:**

- Epilepsy and other CNS disease; gastric ulcer, **pregnancy**

Rimantadine

- Methyl derivative of amantadine
- **More potent, longer acting** ($t_{1/2}$ 30 hours) and better tolerated
- Side effects is **lower**
- Oral bioavailability is higher
- Metabolites are excreted in urine

Oseltamivir

- Influenza A (**amantadine** sensitive as well as resistant), H5N1 (**bird flu**), nH1N1 (**swine flu**) strains and **influenza B**
- An ester **prodrug**; rapidly and nearly completely **hydrolysed** during absorption in intestine and by liver to the active form **oseltamivir carboxylate** (an oral bioavailability of ~ 80%)
- Active **metabolite** is excreted **unchanged** by the kidney
- $t_{1/2}$ of 6–10 hours

Oseltamivir

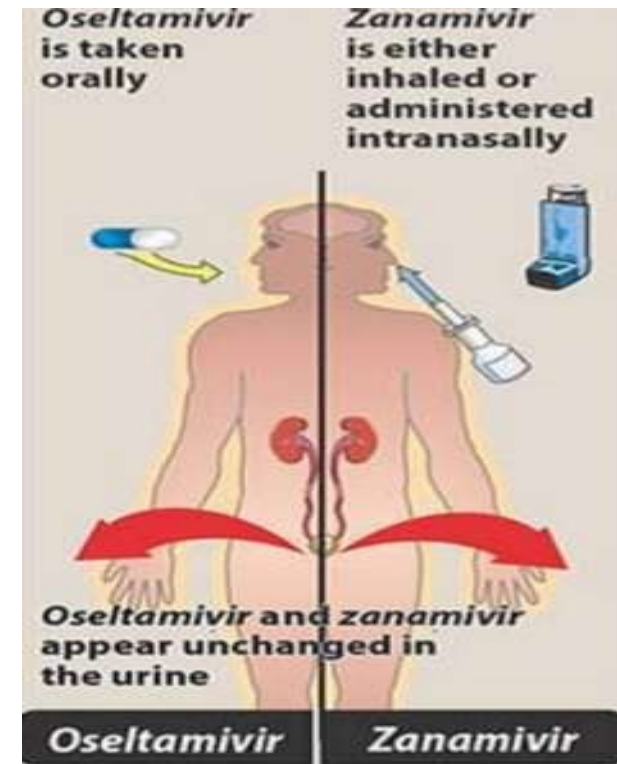
❖ Side effects:

- Nausea and abdominal pain (gastric irritation)
- Headache, weakness, sadness, diarrhoea, cough, and insomnia
- Skin reactions

ZANAMIVIR

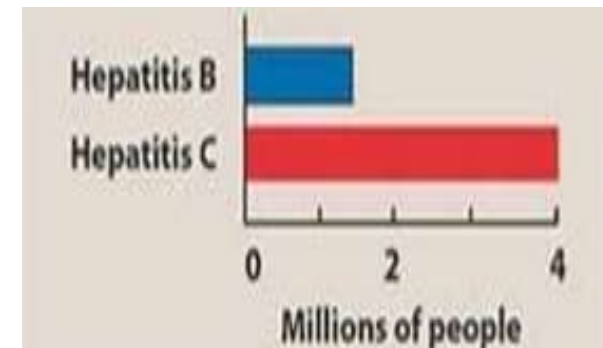
- **Influenza A** (including amantadine-resistant, nH1N1, H5N1 strains) and **influenza B** virus neuraminidase inhibitor
- Low oral bioavailability; $t_{1/2}$ of 2–5 hours

❖ Contraindication: Asthmatics (bronchospasms)



4. Anti-Hepatitis Virus/ Nonselective Antiviral Drugs

- Hepatitis A, B, C, D, and E viruses replicate in and destruct hepatocytes
- Hepatitis B and C are the most common causes of **chronic hepatitis, cirrhosis, and hepatocellular carcinoma**
- Hepatitis B virus (HBV) is a DNA virus, can integrate into host chromosomal DNA to establish permanent infection
- Hepatitis C virus (HCV) is a RNA virus



Lamivudine

- An inhibitor of both **hepatitis B virus (HBV) DNA polymerase** and **human immunodeficiency virus (HIV) reverse transcriptase**
- First line drug for **chronic hepatitis B**
- Chronic treatment  decreased plasma HBV DNA levels, improved biochemical markers, and reduced hepatic inflammation
- ❖ **Mechanism of action:**
 - It must be **phosphorylated by host cellular enzymes** to the **triphosphate** (active) form.
 - This compound **competitively inhibits HBV DNA polymerase** at concentrations that have negligible effects on host DNA polymerase

Adefovir dipivoxil

❖ Mechanism of action:

- It is phosphorylated to adefovir **diphosphate** → incorporated into viral DNA → **termination** of further DNA synthesis → prevents viral replication
- Both **decreased viral load and improved liver function**
- Indicated in **chronic hepatitis B**, also in **lamivudine-resistant cases and in concurrent HIV infection**
- Its **plasma $t_{1/2}$ is 7 hours**; intracellular $t_{1/2}$ of the diphosphate is upto **18 hours**.

Adefovir dipivoxil

❖ **Adverse effects:**

- **Sore throat**, headache, weakness, abdominal pain and flu syndrome
- **Nephrotoxicity** (higher doses and in those with preexisting renal insufficiency)
- **Lactic acidosis** (patients receiving anti-HIV drugs)

Ribavirin

- **Broad-spectrum antiviral activity**
- **Influenza A and B, respiratory syncytial virus** (in children only) and many other DNA and double stranded RNA viruses
- **Oral ribavirin** is commonly used in **chronic hepatitis C**
- ❖ **Mechanism of action:**
 - Its **mono-** and **triphosphate** derivatives generated intracellularly inhibit GTP and viral RNA synthesis

Interferon

❖ Uses:

- Chronic hepatitis B
- Chronic hepatitis C
- AIDS-related Kaposi's sarcoma
- *H. simplex*, *H. zoster* and CMV

<i>Interferon-α</i>	<i>Interferon-β</i>	<i>Interferon-γ</i>
Chronic hepatitis B and C	Relapsing-remitting multiple sclerosis	Chronic granulomatous disease
Genital warts caused by papilloma-virus		
Leukemia, hairy-cell Leukemia, chronic myelogenous		
Kaposi's sarcoma		

Interferon

❖ Adverse effects:

- **Flu-like symptoms:** fatigue, aches and pains, malaise, fever, dizziness, anorexia, nausea, taste and visual disturbances develop few hours after each injection, but become milder later
- **Neurotoxicity:** numbness, neuropathy, altered behaviour, mental depression, tremor, sleepiness, rarely convulsions
- **Myelosuppression:** dose dependent neutropenia, thrombocytopenia
- Thyroid dysfunction (hypo as well as hyper)
- Hypotension, transient arrhythmias, alopecia and liver dysfunction.

Thank you!!